

20. Let p be an odd prime. Prove that if P is a non-cyclic p -group then P contains a normal subgroup U with $U \cong Z_p \times Z_p$. Deduce that for odd primes p a p -group that contains a unique subgroup of order p is cyclic. (For $p = 2$ it is a theorem that the generalized quaternion groups Q_{2^n} are the only non-cyclic 2-groups which contain a unique subgroup of order 2). [Proceed by induction on $|P|$. Let Z be a subgroup of order p in $Z(P)$ and let $\bar{P} = P/Z$. If $\bar{P}$ is cyclic then P is abelian by Exercise 36 in Section 3.1 — show the result is true for abelian groups. When $\bar{P}$ is not cyclic use induction to produce a normal subgroup $\bar{H}$ of $\bar{P}$ with $\bar{H} \cong Z_p \times Z_p$. Let H be the complete preimage of $\bar{H}$ in P , so $|H| = p^3$. Let $H_0 = \{x \in H \mid x^p = 1\}$ so that H_0 is a characteristic subgroup of H of order p^2 or p^3 by Exercise 9 in Section 4. Show that a suitable subgroup of H_0 gives the desired normal subgroup U .]
21. Let p be an odd prime and let P be a p -group. Prove that if every subgroup of P is normal then P is abelian. (Note that Q_8 is a non-abelian 2-group with this property, so the result is false for $p = 2$.) [Use the preceding exercises and Exercise 15 of Section 4.]
22. Let F be a field let n be a positive integer and let G be the group of upper triangular matrices in $GL_n(F)$ (cf. Exercise 16, Section 2.1)
- Prove that G is the semidirect product $U \rtimes D$ where U is the set of upper triangular matrices with 1's down the diagonal (cf. Exercise 17, Section 2.1) and D is the set of diagonal matrices in $GL_n(F)$.
 - Let $n=2$. Recall that $U \cong F$ and $D \cong F^\times \times F^\times$ (cf. Exercise 11 in Section 3.1). Describe the homomorphism from D into $\text{Aut}(U)$ explicitly in terms of these isomorphisms (i.e., show how each element of $F^\times \times F^\times$ acts as an automorphism on F).
23. Let K and L be groups, let n be a positive integer, let $\rho : K \rightarrow S_n$ be a homomorphism and let H be the direct product of n copies of L . In Exercise 8 of Section 1 an injective homomorphism ψ from S_n into $\text{Aut}(H)$ was constructed by letting the elements of S_n permute the n factors of H . The composition $\psi \circ \rho$ is a homomorphism from G into $\text{Aut}(H)$. The *wreath product* of L by K is the semidirect product $H \rtimes K$ with respect to this homomorphism and is denoted by $L \wr K$ (this wreath product depends on the choice of permutation representation ρ of K — if none is given explicitly, ρ is assumed to be the left regular representation of K).
- Assume K and L are finite groups and ρ is the left regular representation of K . Find $|L \wr K|$ in terms of $|K|$ and $|L|$.
 - Let p be a prime, let $K = L = Z_p$ and let ρ be the left regular representation of K . Prove that $Z_p \wr Z_p$ is a non-abelian group of order p^{p+1} and is isomorphic to a Sylow p -subgroup of S_{p^2} . [The p copies of Z_p whose direct product makes up H may be represented by p disjoint p -cycles; these are cyclically permuted by K .]
24. Let n be an integer > 1 . Prove the following classification: every group of order n is abelian if and only if $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$, where $p_1, \dots, p_r$ are distinct primes, $\alpha_i = 1$ or 2 for all $i \in \{1, \dots, r\}$ and p_i does not divide $p_j^{\alpha_j} - 1$ for all i and j . [See Exercise 56 in Section 4.5.]
25. Let $H(\mathbb{F}_p)$ be the Heisenberg group over the finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ (cf. Exercise 20 in Section 4). Prove that $H(\mathbb{F}_2) \cong D_8$, and that $H(\mathbb{F}_p)$ has exponent p and is isomorphic to the first non-abelian group in Example 7.

Further Topics in Group Theory

6.1 p -GROUPS, NILPOTENT GROUPS, AND SOLVABLE GROUPS

Let p be a prime and let G be a finite group of order $p^a n$, where p does not divide n . Recall that a (finite) p -group is any group whose order is a power of p . Sylow's Theorem shows that p -groups abound as subgroups of G and in order to exploit this phenomenon to unravel the structure of finite groups it will be necessary to establish some basic properties of p -groups. In the next section we shall apply these results in many specific instances.

Before giving the results on p -groups we first recall a definition that has appeared in some earlier exercises.

Definition. A *maximal subgroup* of a group G is a proper subgroup M of G such that there are no subgroups H of G with $M < H < G$.

By order considerations every proper subgroup of a finite group is contained in some maximal subgroup. In contrast, infinite groups may or may not have maximal subgroups. For example, $p\mathbb{Z}$ is a maximal subgroup of $\mathbb{Z}$ whereas $\mathbb{Q}$ (under $+$) has no maximal subgroups (cf. Exercise 16 at the end of this section).

We now collect all the properties of p -groups we shall need into an omnibus theorem:

Theorem 1. Let p be a prime and let P be a group of order p^a , $a \geq 1$. Then

- (1) The center of P is nontrivial: $Z(P) \neq 1$.
- (2) If H is a nontrivial normal subgroup of P then H intersects the center non-trivially: $H \cap Z(P) \neq 1$. In particular, every normal subgroup of order p is contained in the center.
- (3) If H is a normal subgroup of P then H contains a subgroup of order p^b that is normal in P for each divisor p^b of $|H|$. In particular, P has a normal subgroup of order p^b for every $b \in \{0, 1, \dots, a\}$.
- (4) If $H < P$ then $H < N_P(H)$ (i.e., every proper subgroup of P is a proper subgroup of its normalizer in P).
- (5) Every maximal subgroup of P is of index p and is normal in P .

Proof: These results rely ultimately on the class equation and it may be useful for the reader to review Section 4.3.

Part 1 is Theorem 8 of Chapter 4 and is also the special case of part 2 when $H = P$. We therefore begin by proving (2); we shall not quote Theorem 8 of Chapter 4 although the argument that follows is only a slight generalization of the one in Chapter 4. Let H be a nontrivial normal subgroup of P . Recall that for each conjugacy class C of P , either $C \subseteq H$ or $C \cap H = \emptyset$ because H is normal (this easy fact was shown in a remark preceding Theorem 4.12). Pick representatives of the conjugacy classes of P :

$$a_1, a_2, \dots, a_r$$

with $a_1, \dots, a_k \in H$ and $a_{k+1}, \dots, a_r \notin H$. Let C_i be the conjugacy class of a_i in P , for all i . Thus

$$C_i \subseteq H, \quad 1 \leq i \leq k \quad \text{and} \quad C_i \cap H = \emptyset, \quad k+1 \leq i \leq r.$$

By renumbering $a_1, \dots, a_k$ if necessary we may assume $a_1, \dots, a_s$ represent classes of size 1 (i.e., are in the center of P) and $a_{s+1}, \dots, a_k$ represent classes of size > 1 . Since H is the disjoint union of these we have

$$|H| = |H \cap Z(P)| + \sum_{i=s+1}^k \frac{|P|}{|C_P(a_i)|}.$$

Now p divides $|H|$ and p divides each term in the sum $\sum_{i=s+1}^k |P : C_P(a_i)|$ so p divides their difference: $|H \cap Z(P)|$. This proves $H \cap Z(P) \neq 1$. If $|H| = p$, since $H \cap Z(P) \neq 1$ we must have $H \leq Z(P)$. This completes the proof of (2).

Next we prove (3) by induction on a . If $a \leq 1$ or $H = 1$, the result is trivial. Assume therefore that $a > 1$ and $H \neq 1$. By part 2, $H \cap Z(P) \neq 1$ so by Cauchy's Theorem $H \cap Z(P)$ contains a (normal) subgroup Z of order p . Use bar notation to denote passage to the quotient group P/Z . This quotient has order p^{a-1} and $\overline{H} \trianglelefteq \overline{P}$. By induction, for every nonnegative integer b such that p^b divides $|\overline{H}|$ there is a subgroup $\overline{K}$ of $\overline{H}$ of order p^b that is normal in $\overline{P}$. If K is the complete preimage of $\overline{K}$ in P then $|K| = p^{b+1}$. The set of all subgroups of H obtained by this process together with the identity subgroup provides a subgroup of H that is normal in P for each divisor of $|H|$. The second assertion of part 3 is the special case $H = P$. This establishes part 3.

We prove (4) also by induction on $|P|$. If P is abelian then all subgroups of P are normal in P and the result is trivial. We may therefore assume $|P| > p$ (in fact, $|P| > p^2$ by Corollary 4.9). Let H be a proper subgroup of P . Since all elements of $Z(P)$ commute with all elements of P , $Z(P)$ normalizes every subgroup of P . By part 1 we have that $Z(P) \neq 1$. If $Z(P)$ is not contained in H , then H is properly contained in $\langle H, Z(P) \rangle$ and the latter subgroup is contained in $N_P(H)$ so (4) holds. We may therefore assume $Z(P) \leq H$. Use bar notation to denote passage to the quotient $P/Z(P)$. Since $\overline{P}$ has smaller order than P by (1), by induction $\overline{H}$ is properly contained in $N_{\overline{P}}(\overline{H})$. It follows directly from the Lattice Isomorphism Theorem that $N_P(H)$ is the complete preimage in P of $N_{\overline{P}}(\overline{H})$, hence we obtain proper containment of H in its normalizer in this case as well. This completes the induction.

To prove (5) let M be a maximal subgroup of P . By definition, $M < P$ so by part 4, $M < N_P(M)$. By definition of maximality we must therefore have $N_P(M) = P$, i.e., $M \trianglelefteq P$. The Lattice Isomorphism Theorem shows that P/M is a p -group with no proper nontrivial subgroups because M is a maximal subgroup. By part 3, however,

P/M has subgroups of every order dividing $|P/M|$. The only possibility is $|P/M| = p$. This proves (5) and completes the proof of the theorem.

Definition.

- (1) For any (finite or infinite) group G define the following subgroups inductively:

$$Z_0(G) = 1, \quad Z_1(G) = Z(G)$$

and $Z_{i+1}(G)$ is the subgroup of G containing $Z_i(G)$ such that

$$Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G))$$

(i.e., $Z_{i+1}(G)$ is the complete preimage in G of the center of $G/Z_i(G)$ under the natural projection). The chain of subgroups

$$Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$$

is called the *upper central series* of G . (The use of the term “upper” indicates that $Z_i(G) \leq Z_{i+1}(G)$.)

- (2) A group G is called *nilpotent* if $Z_c(G) = G$ for some $c \in \mathbb{Z}$. The smallest such c is called the *nilpotence class* of G .

One of the exercises at the end of this section shows that $Z_i(G)$ is a characteristic (hence normal) subgroup of G for all i . We use this fact freely from now on.

Remarks:

- (1) If G is abelian then G is nilpotent (of class 1, provided $|G| > 1$), since in this case $G = Z(G) = Z_1(G)$. One should think of nilpotent groups as lying between abelian and solvable groups in the hierarchy of structure (recall that solvable groups were introduced in Section 3.4; we shall discuss solvable groups further at the end of this section):

cyclic groups $\subset$ *abelian groups* $\subset$ *nilpotent groups* $\subset$ *solvable groups* $\subset$ *all groups*

(all of the above containments are proper, as we shall verify shortly).

- (2) For any finite group there must, by order considerations, be an integer n such that

$$Z_n(G) = Z_{n+1}(G) = Z_{n+2}(G) = \cdots$$

For example, $Z_n(S_3) = 1$ for all $n \in \mathbb{Z}^+$. Once two terms in the upper central series are the same, the chain stabilizes at that point (i.e., all terms thereafter are equal to these two). For example, if $G = Z_2 \times S_3$,

$$Z(G) = Z_1(G) = Z_2(G) = Z_n(G) \quad \text{has order 2 for all } n.$$

By definition, $Z_n(G)$ is a proper subgroup of G for all n for non-nilpotent groups.

- (3) For infinite groups G it may happen that all $Z_i(G)$ are proper subgroups of G (so G is not nilpotent) but

$$G = \bigcup_{i=0}^{\infty} Z_i(G).$$

Groups for which this holds are called *hypercentral* — they enjoy some (but not all) of the properties of nilpotent groups. While we shall be dealing mainly with finite nilpotent groups, results that do not involve the notion of order, Sylow subgroups etc. also hold for infinite groups. Even for infinite groups one of the main techniques for dealing with nilpotent groups is induction on the nilpotence class.

Proposition 2. Let p be a prime and let P be a group of order p^a . Then P is nilpotent of nilpotence class at most $a - 1$.

Proof: For each $i \geq 0$, $P/Z_i(P)$ is a p -group, so

$$\text{if } |P/Z_i(P)| > 1 \text{ then } Z(P/Z_i(P)) \neq 1$$

by Theorem 1(1). Thus if $Z_i(P) \neq G$ then $|Z_{i+1}(P)| \geq p|Z_i(P)|$ and so $|Z_{i+1}(P)| \geq p^{i+1}$. In particular, $|Z_a(P)| \geq p^a$, so $P = Z_a(P)$. Thus P is nilpotent of class $\leq a$. The only way P could be of nilpotence class exactly equal to a would be if $|Z_i(P)| = p^i$ for all i . In this case, however, $Z_{a-2}(P)$ would have index p^2 in P , so $P/Z_{a-2}(P)$ would be abelian (by Corollary 4.9). But then $P/Z_{a-2}(P)$ would equal its center and so $Z_{a-1}(P)$ would equal P , a contradiction. This proves that the class of P is $\leq a - 1$.

Example

Both D_8 and Q_8 are nilpotent of class 2. More generally, D_{2^n} is nilpotent of class $n - 1$. This can be proved inductively by showing that $|Z(D_{2^n})| = 2$ and $D_{2^n}/Z(D_{2^n}) \cong D_{2^{n-1}}$ for $n \geq 3$ (the details are left as an exercise). If n is not a power of 2, D_{2^n} is not nilpotent (cf. Exercise 10).

We now give some equivalent (and often more workable) characterizations of nilpotence for *finite* groups:

Theorem 3. Let G be a finite group, let $p_1, p_2, \dots, p_s$ be the distinct primes dividing its order and let $P_i \in \text{Syl}_{p_i}(G)$, $1 \leq i \leq s$. Then the following are equivalent:

- (1) G is nilpotent
- (2) if $H < G$ then $H < N_G(H)$, i.e., every proper subgroup of G is a proper subgroup of its normalizer in G
- (3) $P_i \trianglelefteq G$ for $1 \leq i \leq s$, i.e., every Sylow subgroup is normal in G
- (4) $G \cong P_1 \times P_2 \times \dots \times P_s$.

Proof: The proof that (1) implies (2) is the same argument as for p -groups — the only fact we needed was if G is nilpotent then so is $G/Z(G)$ — so the details are omitted (cf. the exercises).

To show that (2) implies (3) let $P = P_i$ for some i and let $N = N_G(P)$. Since $P \trianglelefteq N$, Corollary 4.20 gives that P is characteristic in N . Since $P \text{ char } N \trianglelefteq N_G(N)$ we get that $P \trianglelefteq N_G(N)$. This means $N_G(N) \leq N$ and hence $N_G(N) = N$. By (2) we must therefore have $N = G$, which gives (3).

Next we prove (3) implies (4). For any t , $1 \leq t \leq s$ we show inductively that

$$P_1 P_2 \cdots P_t \cong P_1 \times P_2 \times \cdots \times P_t.$$

Note first that each P_i is normal in G so $P_1 \cdots P_t$ is a subgroup of G . Let H be the product $P_1 \cdots P_{t-1}$ and let $K = P_t$, so by induction $H \cong P_1 \times \cdots \times P_{t-1}$. In particular, $|H| = |P_1| \cdot |P_2| \cdots |P_{t-1}|$. Since $|K| = |P_t|$, the orders of H and K are relatively prime. Lagrange's Theorem implies $H \cap K = 1$. By definition, $P_1 \cdots P_t = HK$, hence Theorem 5.9 gives

$$HK \cong H \times K = (P_1 \times \cdots \times P_{t-1}) \times P_t \cong P_1 \times \cdots \times P_t$$

which completes the induction. Now take $t = s$ to obtain (4).

Finally, to prove (4) implies (1) use Exercise 1 of Section 5.1 to obtain

$$Z(P_1 \times \cdots \times P_s) \cong Z(P_1) \times \cdots \times Z(P_s).$$

By Exercise 14 in Section 5.1,

$$G/Z(G) = (P_1/Z(P_1)) \times \cdots \times (P_s/Z(P_s)).$$

Thus the hypotheses of (4) also hold for $G/Z(G)$. By Theorem 1, if $P_i \neq 1$ then $Z(P_i) \neq 1$, so if $G \neq 1$, $|G/Z(G)| < |G|$. By induction, $G/Z(G)$ is nilpotent, so by Exercise 6, G is nilpotent. This completes the proof.

Note that the first part of the Fundamental Theorem of Finite Abelian Groups (Theorem 5 in Section 5.2) follows immediately from the above theorem (we shall give another proof later as a consequence of the Chinese Remainder Theorem):

Corollary 4. A finite abelian group is the direct product of its Sylow subgroups.

Next we prove a proposition which will be used later to show that the multiplicative group of a finite field is cyclic (without using the Fundamental Theorem of Finite Abelian Groups).

Proposition 5. If G is a finite group such that for all positive integers n dividing its order, G contains at most n elements x satisfying $x^n = 1$, then G is cyclic.

Proof: Let $|G| = p_1^{\alpha_1} \cdots p_s^{\alpha_s}$ and let P_i be a Sylow p_i -subgroup of G for $i = 1, 2, \dots, s$. Since $p_i^{\alpha_i} \mid |G|$ and the $p_i^{\alpha_i}$ elements of P_i are solutions of $x^{p_i^{\alpha_i}} = 1$, by hypothesis P_i must contain *all* solutions to this equation in G . It follows that P_i is the unique (hence normal) Sylow p_i -subgroup of G . By Theorem 3, G is the direct product of its Sylow subgroups. By Theorem 1, each P_i possesses a normal subgroup M_i of index p_i . Since $|M_i| = p_i^{\alpha_i-1}$ and G has at most $p_i^{\alpha_i-1}$ solutions to $x^{p_i^{\alpha_i-1}} = 1$, by Lagrange's Theorem (Corollary 9, Section 3.2) M contains all elements x of G satisfying $x^{p_i^{\alpha_i-1}} = 1$. Thus any element of P_i not contained in M_i satisfies $x^{p_i^{\alpha_i}} = 1$ but $x^{p_i^{\alpha_i-1}} \neq 1$, i.e., x is an element of order $p_i^{\alpha_i}$. This proves P_i is cyclic for all i , so G is the direct product of cyclic groups of relatively prime order, hence is cyclic.

The next proposition is called Frattini's Argument. We shall apply it to give another characterization of finite nilpotent groups. It will also be a valuable tool in the next section.

Proposition 6. (*Frattini's Argument*) Let G be a finite group, let H be a normal subgroup of G and let P be a Sylow p -subgroup of H . Then $G = HN_G(P)$ and $|G : H|$ divides $|N_G(P)|$.

Proof: By Corollary 3.15, $HN_G(P)$ is a subgroup of G and $HN_G(P) = N_G(P)H$ since H is a normal subgroup of G . Let $g \in G$. Since $P^g \leq H^g = H$, both P and P^g are Sylow p -subgroups of H . By Sylow's Theorem applied in H , there exists $x \in H$ such that $P^g = P^x$. Thus $gx^{-1} \in N_G(P)$ and so $g \in N_G(P)x$. Since g was an arbitrary element of G , this proves $G = N_G(P)H$.

Apply the Second Isomorphism Theorem to $G = N_G(P)H$ to conclude that

$$|G : H| = |N_G(P) : N_G(P) \cap H|$$

so $|G : H|$ divides $|N_G(P)|$, completing the proof.

Proposition 7. A finite group is nilpotent if and only if every maximal subgroup is normal.

Proof: Let G be a finite nilpotent group and let M be a maximal subgroup of G . As in the proof of Theorem 1, since $M < N_G(M)$ (by Theorem 3(2)) maximality of M forces $N_G(M) = G$, i.e., $M \trianglelefteq G$.

Conversely, assume every maximal subgroup of the finite group G is normal. Let P be a Sylow p -subgroup of G . We prove $P \trianglelefteq G$ and conclude that G is nilpotent by Theorem 3(3). If P is not normal in G let M be a maximal subgroup of G containing $N_G(P)$. By hypothesis, $M \trianglelefteq G$ hence by Frattini's Argument $G = MN_G(P)$. Since $N_G(P) \leq M$ we have $MN_G(P) = M$, a contradiction. This establishes the converse.

Commutators and the Lower Central Series

For the sake of completeness we include the definition of the *lower central series* of a group and state its relation to the upper central series. Since we shall not be using these results in the future, the proofs are left as (straightforward) exercises.

Recall that the commutator of two elements x, y in a group G is defined as

$$[x, y] = x^{-1}y^{-1}xy,$$

and the commutator of two subgroups H and K of G is

$$[H, K] = \langle [h, k] \mid h \in H, k \in K \rangle.$$

Basic properties of commutators and the commutator subgroup were established in Section 5.4.

Definition. For any (finite or infinite) group G define the following subgroups inductively:

$$G^0 = G, \quad G^1 = [G, G] \quad \text{and} \quad G^{i+1} = [G, G^i].$$

The chain of groups

$$G^0 \geq G^1 \geq G^2 \geq \dots$$

is called the *lower central series* of G . (The term “lower” indicates that $G^i \geq G^{i+1}$.)

As with the upper central series we include in the exercises at the end of this section the verification that G^i is a characteristic subgroup of G for all i . The next theorem shows the relation between the upper and lower central series of a group.

Theorem 8. A group G is nilpotent if and only if $G^n = 1$ for some $n \geq 0$. More precisely, G is nilpotent of class c if and only if c is the smallest nonnegative integer such that $G^c = 1$. If G is nilpotent of class c then

$$Z_i(G) \leq G^{c-i-1} \leq Z_{i+1}(G) \quad \text{for all } i \in \{0, 1, \dots, c-1\}.$$

Proof: This is proved by a straightforward induction on the length of either the upper or lower central series.

The terms of the upper and lower central series do not necessarily coincide in general although in some groups this does occur.

Remarks:

- (1) If G is abelian, we have already seen that $G' = G^1 = 1$ so the lower central series terminates in the identity after one term.
- (2) As with the upper central series, for any finite group there must, by order considerations, be an integer n such that

$$G^n = G^{n+1} = G^{n+2} = \dots$$

For non-nilpotent groups, G^n is a nontrivial subgroup of G . For example, in Section 5.4 we showed that $S_3' = S_3^1 = A_3$. Since S_3 is not nilpotent, we must have $S_3^2 = A_3$. In fact

$$(123) = [(12), (132)] \in [S_3, S_3^1] = S_3^2.$$

Once two terms in the lower central series are the same, the chain stabilizes at that point i.e., all terms thereafter are equal to these two. Thus $S_3^i = A_3$ for all $i \geq 2$. Note that S_3 is an example where the lower central series has two distinct terms whereas all terms in the upper central series are equal to the identity (in particular, for non-nilpotent groups these series need not have the same length).

Solvable Groups and the Derived Series

Recall that in Section 3.4 a solvable group was defined as one possessing a series:

$$1 = H_0 \trianglelefteq H_1 \trianglelefteq \dots \trianglelefteq H_s = G$$

such that each factor H_{i+1}/H_i is abelian. We now give another characterization of solvability in terms of a descending series of characteristic subgroups.

Definition. For any group G define the following sequence of subgroups inductively:

$$G^{(0)} = G, \quad G^{(1)} = [G, G] \quad \text{and} \quad G^{(i+1)} = [G^{(i)}, G^{(i)}] \quad \text{for all } i \geq 1.$$

This series of subgroups is called the *derived* or *commutator* series of G .

The terms of this series are also often written as: $G^{(1)} = G'$, $G^{(2)} = G''$, etc. Again it is left as an exercise to show that each $G^{(i)}$ is characteristic in G for all i .

It is important to note that although $G^{(0)} = G^0$ and $G^{(1)} = G^1$, it is not in general true that $G^{(i)} = G^i$. The difference is that the definition of the $i+1^{\text{st}}$ term in the lower central series is the commutator of the i^{th} term with the *whole* group G whereas the $i+1^{\text{st}}$ term in the derived series is the commutator of the i^{th} term with itself. Hence

$$G^{(i)} \leq G^i \quad \text{for all } i$$

and the containment can be proper. For example, in $G = S_3$ we have already seen that $G^1 = G' = A_3$ and $G^2 = [S_3, A_3] = A_3$, whereas $G^{(2)} = [A_3, A_3] = 1$ (A_3 being abelian).

Theorem 9. A group G is solvable if and only if $G^{(n)} = 1$ for some $n \geq 0$.

Proof: Assume first that G is solvable and so possesses a series

$$1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_s = G$$

such that each factor H_{i+1}/H_i is abelian. We prove by induction that $G^{(i)} \leq H_{s-i}$. This is true for $i = 0$, so assume $G^{(i)} \leq H_{s-i}$. Then

$$G^{(i+1)} = [G^{(i)}, G^{(i)}] \leq [H_{s-i}, H_{s-i}].$$

Since H_{s-i}/H_{s-i-1} is abelian, by Proposition 5.7(4), $[H_{s-i}, H_{s-i}] \leq H_{s-i-1}$. Thus $G^{(i+1)} \leq H_{s-i-1}$, which completes the induction. Since $H_0 = 1$ we have $G^{(s)} = 1$.

Conversely, if $G^{(n)} = 1$ for some $n \geq 0$, Proposition 5.7(4) shows that if we take H_i to be $G^{(n-i)}$ then H_i is a normal subgroup of H_{i+1} with abelian quotient, so the derived series itself satisfies the defining condition for solvability of G . This completes the proof.

If G is solvable, the smallest nonnegative n for which $G^{(n)} = 1$ is called the *solvable length* of G . The derived series is a series of shortest length whose successive quotients are abelian and it has the additional property that it consists of subgroups that are characteristic in the *whole* group (as opposed to each just being normal in the *next* in the initial definition of solvability). Its “intrinsic” definition also makes it easier to work with in many instances, as the following proposition (which reproves some results and exercises from Section 3.4) illustrates.

Proposition 10. Let G and K be groups, let H be a subgroup of G and let $\varphi : G \rightarrow K$ be a surjective homomorphism.

- (1) $H^{(i)} \leq G^{(i)}$ for all $i \geq 0$. In particular, if G is solvable, then so is H , i.e., subgroups of solvable groups are solvable (and the solvable length of H is less than or equal to the solvable length of G).

- (2) $\varphi(G^{(i)}) = K^{(i)}$. In particular, homomorphic images and quotient groups of solvable groups are solvable (of solvable length less than or equal to that of the domain group).
- (3) If N is normal in G and both N and G/N are solvable then so is G .

Proof: Part 1 follows from the observation that since $H \leq G$, by definition of commutator subgroups, $[H, H] \leq [G, G]$, i.e., $H^{(1)} \leq G^{(1)}$. Then, by induction,

$$H^{(i)} \leq G^{(i)} \quad \text{for all } i \in \mathbb{Z}^+.$$

In particular, if $G^{(n)} = 1$ for some n , then also $H^{(n)} = 1$. This establishes (1).

To prove (2) note that by definition of commutators,

$$\varphi([x, y]) = [\varphi(x), \varphi(y)]$$

so by induction $\varphi(G^{(i)}) \leq K^{(i)}$. Since φ is surjective, every commutator in K is the image of a commutator in G , hence again by induction we obtain equality for all i . Again, if $G^{(n)} = 1$ for some n then $K^{(n)} = 1$. This proves (2).

Finally, if G/N and N are solvable, of lengths n and m respectively then by (2) applied to the natural projection $\varphi : G \rightarrow G/N$ we obtain

$$\varphi(G^{(n)}) = (G/N)^{(n)} = 1N$$

i.e., $G^{(n)} \leq N$. Thus $G^{(n+m)} = (G^{(n)})^{(m)} \leq N^{(m)} = 1$. Theorem 9 shows that G is solvable, which completes the proof.

Some additional conditions under which finite groups are solvable are the following:

Theorem 11. Let G be a finite group.

- (1) (Burnside) If $|G| = p^a q^b$ for some primes p and q , then G is solvable.
- (2) (Philip Hall) If for every prime p dividing $|G|$ we factor the order of G as $|G| = p^a m$ where $(p, m) = 1$, and G has a subgroup of order m , then G is solvable (i.e., if for all primes p , G has a subgroup whose index equals the order of a Sylow p -subgroup, then G is solvable — such subgroups are called Sylow p -complements).
- (3) (Feit–Thompson) If $|G|$ is odd then G is solvable.
- (4) (Thompson) If for every pair of elements $x, y \in G$, $\langle x, y \rangle$ is a solvable group, then G is solvable.

We shall prove Burnside's Theorem in Chapter 19 and deduce Philip Hall's generalization of it. As mentioned in Section 3.5, the proof of the Feit–Thompson Theorem takes 255 pages. Thompson's Theorem was first proved as a consequence of a 475 page paper (that in turn relies ultimately on the Feit–Thompson Theorem).

A Proof of the Fundamental Theorem of Finite Abelian Groups

We sketch a group-theoretic proof of the result that every finite abelian group is a direct product of cyclic groups (i.e., Parts 1 and 2 of Theorem 5, Section 5.2) — the Classification of Finitely Generated Abelian Groups (Theorem 3, Section 5.2) will be derived as a consequence of a more general theorem in Chapter 12.

By Corollary 4 it suffices to prove that for p a prime, any abelian p -group is a direct product of cyclic groups (the divisibility condition in Theorem 5.5 is trivially achieved by reordering factors). Let A be an abelian p -group. We proceed by induction on $|A|$.

If E is an elementary abelian p -group (i.e., $x^p = 1$ for all $x \in E$), we first prove the following result:

for any $x \in E$, there exists $M \leq E$ with $E = M \times \langle x \rangle$.

If $x = 1$, let $M = E$. Otherwise let M be a subgroup of E of maximal order subject to the condition that x not be an element of M . If M is not of index p in E , let $\bar{E} = E/M$. Then $\bar{E}$ is elementary abelian and there exists $\bar{y} \in \bar{E} - \langle \bar{x} \rangle$. Since $\bar{y}$ has order p , we also have $\bar{x} \notin \langle \bar{y} \rangle$. The complete preimage of $\langle \bar{y} \rangle$ in E is a subgroup of E that does not contain x and whose order is larger than the order of M , contrary to the choice of M . This proves $|E : M| = p$, hence

$$E = M\langle x \rangle \quad \text{and} \quad M \cap \langle x \rangle = 1.$$

By the recognition theorem for direct products, Theorem 5.9, $E = M \times \langle x \rangle$, as asserted.

Now let $\varphi : A \rightarrow A$ be defined by $\varphi(x) = x^p$ (see Exercise 7, Section 5.2). Then φ is a homomorphism since A is abelian. Denote the kernel of φ by K and denote the image of φ by H . By definition $K = \{x \in A \mid x^p = 1\}$ and H is the subgroup of A consisting of p^{th} powers. Note that both K and A/H are elementary abelian. By the First Isomorphism Theorem

$$|A : H| = |K|.$$

By induction,

$$\begin{aligned} H &= \langle h_1 \rangle \times \cdots \times \langle h_r \rangle \\ &\cong Z_{p^{\alpha_1}} \times \cdots \times Z_{p^{\alpha_r}} \quad \alpha_i \geq 1, \quad i = 1, 2, \dots, r. \end{aligned}$$

By definition of φ , there exist elements $g_i \in A$ such that $g_i^p = h_i$, $1 \leq i \leq r$. Let $A_0 = \langle g_1, \dots, g_r \rangle$. It is an exercise to see that

- (a) $A_0 = \langle g_1 \rangle \times \cdots \times \langle g_r \rangle$,
- (b) $A_0/H = \langle g_1H \rangle \times \cdots \times \langle g_rH \rangle$ is elementary abelian of order p^r , and
- (c) $H \cap K = \langle h_1^{p^{\alpha_1-1}} \rangle \times \cdots \times \langle h_r^{p^{\alpha_r-1}} \rangle$ is elementary abelian of order p^r .

If K is contained in H , then $|K| = |K \cap H| = p^r = |A_0 : H|$. In this case by comparing orders we see that $A_0 = A$ and the theorem is proved. Assume therefore that K is not a subgroup of H and use the bar notation to denote passage to the quotient group A/H . Let $x \in K - H$, so $|\bar{x}| = |x| = p$. By the initial remark of the proof applied to the elementary abelian p -group $E = \bar{A}$, there is a subgroup $\bar{M}$ of $\bar{A}$ such that

$$\bar{A} = \bar{M} \times \langle \bar{x} \rangle.$$

If M is the complete preimage in A of $\bar{M}$, then since x has order p and $x \notin M$, we have $\langle x \rangle \cap M = 1$. By the recognition theorem for direct products,

$$A = M \times \langle x \rangle.$$

By induction, M is a direct product of cyclic groups, hence so is A . This completes the proof.

The uniqueness of the decomposition of a finite abelian group into a direct product of cyclic groups (Part 3 of Theorem 5.5) can also be proved by induction using the p^{th} -power map (i.e., using Exercise 7, Section 5.2). This is essentially the procedure we follow in Section 12.1 for the uniqueness part of the proof of the Fundamental Theorem of Finitely Generated Abelian Groups.

EXERCISES

1. Prove that $Z_i(G)$ is a characteristic subgroup of G for all i .
2. Prove Parts 2 and 4 of Theorem 1 for G a finite nilpotent group, not necessarily a p -group.
3. If G is finite prove that G is nilpotent if and only if it has a normal subgroup of each order dividing $|G|$, and is cyclic if and only if it has a unique subgroup of each order dividing $|G|$.
4. Prove that a maximal subgroup of a finite nilpotent group has prime index.
5. Prove Parts 2 and 4 of Theorem 1 for G an infinite nilpotent group.
6. Show that if $G/Z(G)$ is nilpotent then G is nilpotent.
7. Prove that subgroups and quotient groups of nilpotent groups are nilpotent (your proof should work for infinite groups). Give an explicit example of a group G which possesses a normal subgroup H such that both H and G/H are nilpotent but G is not nilpotent.
8. Prove that if p is a prime and P is a non-abelian group of order p^3 then $|Z(P)| = p$ and $P/Z(P) \cong Z_p \times Z_p$.
9. Prove that a finite group G is nilpotent if and only if whenever $a, b \in G$ with $(|a|, |b|) = 1$ then $ab = ba$. [Use Part 4 of Theorem 3.]
10. Prove that D_{2^n} is nilpotent if and only if n is a power of 2. [Use Exercise 9.]
11. Give another proof of Proposition 5 under the additional assumption that G is abelian by invoking the Fundamental Theorem of Finite Abelian Groups.
12. Find the upper and lower central series for A_4 and S_4 .
13. Find the upper and lower central series for A_n and S_n , $n \geq 5$.
14. Prove that G^i is a characteristic subgroup of G for all i .
15. Prove that $Z_i(D_{2^n}) = D_{2^n}^{n-1-i}$.
16. Prove that $\mathbb{Q}$ has no maximal subgroups. [Recall Exercise 21, Section 3.2.]
17. Prove that $G^{(i)}$ is a characteristic subgroup of G for all i .
18. Show that if G'/G'' and G''/G''' are both cyclic then $G'' = 1$. [You may assume $G''' = 1$. Then G/G'' acts by conjugation on the cyclic group G'' .]
19. Show that there is no group whose commutator subgroup is isomorphic to S_4 . [Use the preceding exercise.]
20. Let p be a prime, let P be a p -subgroup of the finite group G , let N be a normal subgroup of G whose order is relatively prime to p and let $\overline{G} = G/N$. Prove the following:
 - (a) $N_{\overline{G}}(\overline{P}) = \overline{N_G(P)}$ [Use Frattini's Argument.]
 - (b) $C_{\overline{G}}(\overline{P}) = \overline{C_G(P)}$. [Use part (a).]

For any group G the *Frattini subgroup* of G (denoted by $\Phi(\hat{G})$) is defined to be the intersection of all the maximal subgroups of G (if G has no maximal subgroups, set $\Phi(G) = G$). The next

few exercises deal with this important subgroup.

21. Prove that $\Phi(G)$ is a characteristic subgroup of G .
22. Prove that if $N \leq G$ then $\Phi(N) \leq \Phi(G)$. Give an explicit example where this containment does not hold if N is not normal in G .
23. Compute $\Phi(S_3)$, $\Phi(A_4)$, $\Phi(S_4)$, $\Phi(A_5)$ and $\Phi(S_5)$.
24. Say an element x of G is a *nongenerator* if for every proper subgroup H of G , $\langle x, H \rangle$ is also a proper subgroup of G . Prove that $\Phi(G)$ is the set of nongenerators of G (here $|G| > 1$).
25. Let G be a finite group. Prove that $\Phi(G)$ is nilpotent. [Use Frattini's Argument to prove that every Sylow subgroup of $\Phi(G)$ is normal in G .]
26. Let p be a prime, let P be a finite p -group and let $\bar{P} = P/\Phi(P)$.
 - (a) Prove that $\bar{P}$ is an elementary abelian p -group. [Show that $P' \leq \Phi(P)$ and that $x^p \in \Phi(P)$ for all $x \in P$.]
 - (b) Prove that if N is any normal subgroup of P such that P/N is elementary abelian then $\Phi(P) \leq N$. State this (universal) property in terms of homomorphisms and commutative diagrams.
 - (c) Let $\bar{P}$ be elementary abelian of order p^r (by (a)). Deduce from Exercise 24 that if $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_r$ are any basis for the r -dimensional vector space $\bar{P}$ over $\mathbb{F}_p$ and if x_i is any element of the coset $\bar{x}_i$, then $P = \langle x_1, x_2, \dots, x_r \rangle$. Show conversely that if $y_1, y_2, \dots, y_s$ is any set of generators for P , then $s \geq r$ (you may assume that every minimal generating set for an r -dimensional vector space has r elements, i.e., every basis has r elements). Deduce *Burnside's Basis Theorem*: a set $y_1, \dots, y_s$ is a minimal generating set for P if and only if $\bar{y}_1, \dots, \bar{y}_s$ is a basis of $\bar{P} = P/\Phi(P)$. Deduce that any minimal generating set for P has r elements.
 - (d) Prove that if $P/\Phi(P)$ is cyclic then P is cyclic. Deduce that if P/P' is cyclic then so is P .
 - (e) Let σ be any automorphism of P of prime order q with $q \neq p$. Show that if σ fixes the coset $x\Phi(P)$ then σ fixes some element of this coset (note that since $\Phi(P)$ is characteristic in P every automorphism of P induces an automorphism of $P/\Phi(P)$). [Use the observation that σ acts a permutation of order 1 or q on the p^a elements in the coset $x\Phi(P)$.]
 - (f) Use parts (e) and (c) to deduce that every nontrivial automorphism of P of order prime to p induces a nontrivial automorphism on $P/\Phi(P)$. Deduce that any group of automorphisms of P which has order prime to p is isomorphic to a subgroup of $\text{Aut}(\bar{P}) = \text{GL}_r(\mathbb{F}_p)$.
27. Generalize part (d) of the preceding exercise as follows: let p be a prime, let P be a p -group and let $\bar{P} = P/\Phi(P)$ be elementary abelian of order p^r . Prove that P has exactly $\frac{p^r - 1}{p - 1}$ maximal subgroups. [Since every maximal subgroup of P contains $\Phi(P)$, the maximal subgroups of P are, by the Lattice Isomorphism Theorem, in bijective correspondence with the maximal subgroups of the elementary abelian group $\bar{P}$. It therefore suffices to show that the number of maximal subgroups of an elementary abelian p -group of order p^r is as stated above. One way of doing this is to use the result that an abelian group is isomorphic to its dual group (cf. Exercise 14 in Section 5.2) so the number of subgroups of index p equals the number of subgroups of order p .]
28. Prove that if p is a prime and $P = Z_p \times Z_{p^2}$ then $|\Phi(P)| = p$ and $P/\Phi(P) \cong Z_p \times Z_p$. Deduce that P has $p + 1$ maximal subgroups.

29. Prove that if p is a prime and P is a non-abelian group of order p^3 then $\Phi(P) = Z(P)$ and $P/\Phi(P) \cong Z_p \times Z_p$. Deduce that P has $p + 1$ maximal subgroups.
30. Let p be an odd prime, let $P_1 = Z_p \times Z_{p^2}$ and let P_2 be the non-abelian group of order p^3 which has an element of order p^2 . Prove that P_1 and P_2 have the same lattice of subgroups.
31. For any group G a *minimal normal subgroup* is a normal subgroup M of G such that the only normal subgroups of G which are contained in M are 1 and M . Prove that every minimal normal subgroup of a finite solvable group is an elementary abelian p -group for some prime p . [If M is a minimal normal subgroup of G , consider its characteristic subgroups: M' and $\langle x^p \mid x \in M \rangle$.]
32. Prove that every maximal subgroup of a finite solvable group has prime power index. [Let H be a maximal subgroup of G and let M be a minimal normal subgroup of G — cf. the preceding exercise. Apply induction to G/M and consider separately the two cases: $M \leq H$ and $M \not\leq H$.]
33. Let π be any set of primes. A subgroup H of a finite group is called a *Hall π -subgroup* of G if the only primes dividing $|H|$ are in the set π and $|H|$ is relatively prime to $|G : H|$. (Note that if $\pi = \{p\}$, Hall π -subgroups are the same as Sylow p -subgroups. Hall subgroups were introduced in Exercise 10 of Section 3.3). Prove the following generalization of Sylow's Theorem for solvable groups: if G is a finite solvable group then for every set π of primes, G has a Hall π -subgroup and any two Hall π -subgroups (for the same set π) are conjugate in G . [Fix π and proceed by induction on $|G|$, proving both existence and conjugacy at once. Let M be a minimal normal subgroup of G , so M is a p -group for some prime p . If $p \in \pi$, apply induction to G/M . If $p \notin \pi$, reduce to the case $|G| = p^\alpha n$, where $p^\alpha = |M|$ and n is the order of a Hall π -subgroup of G . In this case let N/M be a minimal normal subgroup of G/M , so N/M is a q -group for some prime $q \neq p$. Let $Q \in \text{Syl}_q(N)$. If $Q \trianglelefteq G$ argue as before with Q in place of M . If Q is not normal in G , use Frattini's Argument to show $N_G(Q)$ is a Hall π -subgroup of G and establish conjugacy in this case too.]

The following result shows how to produce normal p -subgroups of some groups on which the elements of order prime to p act faithfully by conjugation. Exercise 26(f) then applies to restrict these actions and give some information about the structure of the group.

34. Let p be a prime dividing the order of the finite solvable group G . Assume G has no nontrivial normal subgroups of order prime to p . Let P be the largest normal p -subgroup of G (cf. Exercise 37, Section 4.5). Note that Exercise 31 above shows that $P \neq 1$. Prove that $C_G(P) \leq P$, i.e., $C_G(P) = Z(P)$. [Let $N = C_G(P)$ and use the preceding exercise to show $N = Z(P) \times H$ for some Hall π -subgroup H of N — here π is the set of all prime divisors of $|N|$ except for p . Show $H \trianglelefteq G$ to obtain the desired conclusion: $H = 1$.]
35. Prove that if G is a finite group in which every proper subgroup is nilpotent, then G is solvable. [Show that a minimal counterexample is simple. Let M and N be distinct maximal subgroups chosen with $|M \cap N|$ as large as possible and apply Part 2 of Theorem 3 to show that $M \cap N = 1$. Now apply the methods of Exercise 53 in Section 4.5.]
36. Let p be a prime, let V be a nonzero finite dimensional vector space over the field of p elements and let φ be an element of $GL(V)$ of order a power of p (i.e., V is a nontrivial elementary abelian p -group and φ is an automorphism of V of p -power order). Prove that there is some nonzero element $v \in V$ such that $\varphi(v) = v$, i.e., φ has a nonzero fixed point on V .
37. Let V be a finite dimensional vector space over the field of 2 elements and let φ be an element of $GL(V)$ of order 2. (i.e., V is a nontrivial elementary abelian 2-group and φ is an

automorphism of V of order 2). Prove that the map $v \mapsto v + \varphi(v)$ is a homomorphism from V to itself. Show that every element in the image of this map is fixed by φ . Deduce that the subspace of elements of V which are fixed by φ has dimension $\geq \frac{1}{2}(\text{dimension } V)$. (Note that if G is the semidirect product of V with $\langle \varphi \rangle$, where $V \trianglelefteq G$ and φ acts by conjugation on V by sending each $v \in V$ to $\varphi(v)$, then the fixed points of φ on V are $C_V(\varphi)$ and the above map is simply the commutator map: $v \mapsto [v, \varphi]$. In this terminology the problem is to show that $|C_V(\varphi)|^2 \geq |V|$.)

38. Use the preceding exercise to prove that if P is a 2-group which has a cyclic center and M is a subgroup of index 2 in P , then the center of M has rank ≤ 2 . [The group G/M of order 2 acts by conjugation on the $\mathbb{F}_2$ vector space: $\{z \in Z(M) \mid z^2 = 1\}$ and the fixed points of this action are in the center of P .]

6.2 APPLICATIONS IN GROUPS OF MEDIUM ORDER

The purpose of this section is to work through a number of examples which illustrate many of the techniques we have developed. These examples use Sylow's Theorems extensively and demonstrate how they are applied in the study of finite groups. Motivated by the Hölder Program we address primarily the problem of showing that for certain n every group of order n has a proper, nontrivial normal subgroup (i.e., there are no simple groups of order n). In most cases we shall stop once this has been accomplished. However readers should be aware that in the process of achieving this result we shall already have determined a great deal of information about arbitrary groups of given order n for the n that we consider. This information could be built upon to classify groups of these orders (but in general this requires techniques beyond the simple use of semidirect products to construct groups).

Since for p a prime we have already proved that there are no simple p -groups (other than the cyclic group of order p , Z_p) and since the structure of p -groups can be very complicated (recall the table in Section 5.3), we shall not study the structure of p -groups explicitly. Rather, the theory of p -groups developed in the preceding section will be applied to subgroups of groups of non-prime-power order.

Finally, for certain n (e.g., 60, 168, 360, 504,...) there do exist simple groups of order n so, of course, we cannot force every group of these orders to be nonsimple. As in Section 4.5 we can, in certain cases, prove there is a unique simple group of order n and unravel some of its internal structure (Sylow numbers, etc.). We shall study simple groups of order 168 as an additional test case. Thus the Sylow Theorems will be applied in a number of different contexts to show how groups of a given order may be manipulated.

We shall end this section with some comments on the existence problem for groups, particularly for finite simple groups.

For $n < 10000$ there are 60 odd, non-prime-power numbers for which the congruence conditions of Sylow's Theorems do *not* force at least one of the Sylow subgroups to be normal i.e., n_p can be > 1 for all primes $p \mid n$ (recall that n_p denotes the number of Sylow p -subgroups). For example, no numbers of the form pq , where p and q are distinct primes occur in our list by results of Section 4.5. In contrast, for even numbers < 500 there are already 46 candidates for orders of simple groups (the congruence

conditions allow many more possibilities). Many of our numerical examples arise from these lists of numbers and we often use odd numbers because the Sylow congruence conditions allow fewer values for n_p . The purpose of these examples is to illustrate the use of the results we have proved. Many of these examples can be dealt with by more advanced techniques (for example, the Feit–Thompson Theorem proves that there are *no* simple groups of odd composite order).

As we saw in the case $n = 30$ in Section 4.5, even though Sylow’s Theorem permitted $n_5 = 6$ and $n_3 = 10$, further examination showed that any group of order 30 must have both $n_5 = 1$ and $n_3 = 1$. Thus the congruence part of Sylow’s Theorem is a sufficient but by no means necessary condition for normality of a Sylow subgroup. For many n (e.g., $n = 120$) we can prove that there are no simple groups of order n , so there is a nontrivial normal subgroup but this subgroup may not be a Sylow subgroup. For example, S_5 and $SL_2(\mathbb{F}_5)$ both have order 120. The group S_5 has a unique nontrivial proper normal subgroup of order 60 (A_5) and $SL_2(\mathbb{F}_5)$ has a unique nontrivial proper normal subgroup of order 2 ($Z(SL_2(\mathbb{F}_5)) \cong Z_2$), neither of which is a Sylow subgroup. Our techniques for producing normal subgroups must be flexible enough to cover such diverse possibilities. In this section we shall examine Sylow subgroups for different primes dividing n , intersections of Sylow subgroups, normalizers of p -subgroups and many other less obvious subgroups. The elementary methods we outline are by no means exhaustive, even for groups of “medium” order.

Some Techniques

Before listing some techniques for producing normal subgroups in groups of a given (“medium”) order we note that in all the problems where one deals with groups of order n , for some specific n , it is first necessary to factor n into prime powers and then to compute the permissible values of n_p , for all primes p dividing n . We emphasize the need to be comfortable computing mod p when carrying out the last step. The techniques we describe may be listed as follows:

- (1) Counting elements.
- (2) Exploiting subgroups of small index.
- (3) Permutation representations.
- (4) Playing p -subgroups off against each other for different primes p .
- (5) Studying normalizers of intersections of Sylow p -subgroups.

Counting Elements

Let G be a group of order n , let p be a prime dividing n and let $P \in \text{Syl}_p(G)$. If $|P| = p$, then every nonidentity element of P has order p and every element of G of order p lies in some conjugate of P . By Lagrange’s Theorem distinct conjugates of P intersect in the identity, hence in this case the number of elements of G of order p is $n_p(p - 1)$.

If Sylow p -subgroups for different primes p have prime order and we assume none of these is normal, we can sometimes show that the number of elements of prime order is $> |G|$. This contradiction would show that at least one of the n_p ’s must be 1 (i.e., some Sylow subgroup is normal in G).

This is the argument we used (in Section 4.5) to prove that there are no simple

groups of order 30. For another example, suppose $|G| = 105 = 3 \cdot 5 \cdot 7$. If G were simple, we must have $n_3 = 7$, $n_5 = 21$ and $n_7 = 15$. Thus

$$\begin{array}{rcl} \text{the number of elements of order 3 is } 7 \cdot 2 & = & 14 \\ \text{the number of elements of order 5 is } 21 \cdot 4 & = & 84 \\ \text{the number of elements of order 7 is } 15 \cdot 6 & = & 90 \\ \hline \text{the number of elements of prime order is } 188 & > & |G|. \end{array}$$

Sometimes counting elements of prime order does not lead to too many elements. However, there may be so few elements remaining that there must be a normal subgroup involving these elements. This was (in essence) the technique used in Section 4.5 to show that in a group of order 12 either $n_2 = 1$ or $n_3 = 1$. This technique works particularly well when G has a Sylow p -subgroup P of order p such that $N_G(P) = P$. For example, let $|G| = 56$. If G were simple, the only possibility for the number of Sylow 7-subgroups is 8, so

$$\text{the number of elements of order 7 is } 8 \cdot 6 = 48.$$

Thus there are $56 - 48 = 8$ elements remaining in G . Since a Sylow 2-subgroup contains 8 elements (none of which have order 7), there can be at most one Sylow 2-subgroup, hence G has a normal Sylow 2-subgroup.

Exploiting Subgroups of Small Index

Recall that the results of Section 4.2 show that if G has a subgroup H of index k , then there is a homomorphism from G into the symmetric group S_k whose kernel is contained in H . If $k > 1$, this kernel is a proper normal subgroup of G and if we are trying to prove that G is not simple, we may, by way of contradiction, assume that this kernel is the identity. Then, by the First Isomorphism Theorem, G is isomorphic to a subgroup of S_k . In particular, the order of G divides $k!$. This argument shows that if k is the smallest integer with $|G|$ dividing $k!$ for a finite simple group G then G contains no proper subgroups of index less than k . This smallest permissible index k should be calculated at the outset of the study of groups of a given order n . In the examples we consider this is usually quite easy: n will often factor as

$$p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s} \quad \text{with} \quad p_1 < p_2 < \cdots < p_s$$

and α_s is usually equal to 1 or 2 in our examples. In this case the minimal index of a proper subgroup will have to be at least p_s (respectively $2p_s$) and this is often its exact value.

For example, there is no simple group of order 3393, because if $n = 3393 = 3^2 \cdot 13 \cdot 29$, then the minimal index of a proper subgroup is 29 (n does not divide $28!$ because 29 does not divide $28!$). However any simple group of order 3393 must have $n_3 = 13$, so for $P \in \text{Syl}_3(G)$, $N_G(P)$ has index 13, a contradiction.

Permutation Representations

This method is a refinement of the preceding one. As above, if G is a simple group of order n with a proper subgroup of index k , then G is isomorphic to a subgroup of S_k . We may identify G with this subgroup and so assume $G \leq S_k$. Rather than relying only

on Lagrange's Theorem for our contradiction (this was what we did for the preceding technique) we can sometimes show by calculating within S_k that S_k contains no simple subgroup of order n . Two restrictions which may enable one to show such a result are

- (1) if G contains an element or subgroup of a particular order, so must S_k , and
- (2) if $P \in \text{Syl}_p(G)$ and if P is also a Sylow p -subgroup of S_k , then $|N_G(P)|$ must divide $|N_{S_k}(P)|$.

Condition (2) arises frequently when p is a prime, $k = p$ or $p + 1$ and G has a subgroup of index k . In this case p^2 does not divide $k!$, so Sylow p -subgroups of G are also Sylow p -subgroups of S_k . Since now Sylow p -subgroups of S_k are precisely the groups generated by a p -cycle, and distinct Sylow p -subgroups intersect in the identity,

$$\begin{aligned} \text{the no. of Sylow } p\text{-subgroups of } S_k &= \frac{\text{the no. of } p\text{-cycles}}{\text{the no. of } p\text{-cycles in a Sylow } p\text{-subgroup}} \\ &= \frac{k \cdot (k-1) \cdots (k-p+1)}{p(p-1)}. \end{aligned}$$

This number gives the index in S_k of the normalizer of a Sylow p -subgroup of S_k . Thus for $k = p$ or $p + 1$

$$|N_{S_k}(P)| = p(p-1) \quad (k = p \text{ or } k = p + 1)$$

(cf. also the corresponding discussion for centralizers of elements in symmetric groups in Section 4.3 and the last exercises in Section 4.3). This proves, under the above hypotheses, that $|N_G(P)|$ must divide $p(p-1)$.

For example, if G were a simple group of order $396 = 2^2 \cdot 3^2 \cdot 11$, we must have $n_{11} = 12$, so if $P \in \text{Syl}_{11}(G)$, $|G : N_G(P)| = 12$ and $|N_G(P)| = 33$. Since G has a subgroup of index 12, G is isomorphic to a subgroup of S_{12} . But then (considering G as actually contained in S_{12}) $P \in \text{Syl}_{11}(S_{12})$ and $|N_{S_{12}}(P)| = 110$. Since $N_G(P) \leq N_{S_{12}}(P)$, this would imply $33 \mid 110$, clearly impossible, so we cannot have a simple group of order 396.

We can sometimes squeeze a little bit more out of this method by working in A_k rather than S_k . This slight improvement helps only occasionally and only for groups of even order. It is based on the following observations (the first of which we have made earlier in the text).

Proposition 12.

- (1) If G has no subgroup of index 2 and $G \leq S_k$, then $G \leq A_k$.
- (2) If $P \in \text{Syl}_p(S_k)$ for some odd prime p , then $P \in \text{Syl}_p(A_k)$ and $|N_{A_k}(P)| = \frac{1}{2}|N_{S_k}(P)|$.

Proof: The first assertion follows from the Second Isomorphism Theorem: if G is not contained in A_k , then $A_k < GA_k$ so we must have $GA_k = S_k$. But now

$$2 = |S_k : A_k| = |GA_k : A_k| = |G : G \cap A_k|$$

so G has a subgroup, $G \cap A_k$, of index 2.

To prove (2) note that if $P \in \text{Syl}_p(S_k)$, for some odd prime p , by (1) (or order considerations) $P \leq A_k$, hence $P \in \text{Syl}_p(A_k)$ as well. By Frattini's Argument (Proposition 6)

$$S_k = N_{S_k}(P)A_k$$

so, in particular, $N_{S_k}(P)$ is not contained in A_k . This forces $N_{S_k}(P) \cap A_k (= N_{A_k}(P))$ to be a subgroup of index 2 in $N_{S_k}(P)$.

For example, there is no simple group of order 264. Suppose G were a simple group of order $264 = 2^3 \cdot 3 \cdot 11$. We must have $n_{11} = 12$. As usual, G would be isomorphic to a subgroup of S_{12} . Since G is simple (hence contains no subgroup of index 2), $G \leq A_{12}$. Let $P \in \text{Syl}_{11}(G)$. Since $n_{11} = 12 = |G : N_G(P)|$, we have $|N_G(P)| = 22$. As above,

$$|N_{A_{12}}(P)| = \frac{1}{2}|N_{S_{12}}(P)| = \frac{1}{2}11(11 - 1) = 55;$$

however, 22 does not divide 55, a contradiction to $N_G(P) \leq N_{A_{12}}(P)$.

Finally, we emphasize that we have only barely touched upon the combinatorial information available from certain permutation representations. Whenever possible in the remaining examples we shall illustrate other applications of this technique.

Playing p -Subgroups Off Against Each Other for Different Primes p

Suppose p and q are distinct primes such that every group of order pq is cyclic. This is equivalent to $p \nmid q - 1$, where $p < q$. If G has a Sylow q -subgroup Q of order q and $p \mid |N_G(Q)|$, applying Cauchy's Theorem in $N_G(Q)$ gives a group P of order p normalizing Q (note that P need not be a Sylow p -subgroup of G). Thus PQ is a group and if PQ is abelian, we obtain

$$PQ \leq N_G(P) \quad \text{and so} \quad q \mid |N_G(P)|.$$

(A symmetric argument applies if Sylow p -subgroups of G have order p and q divides the order of a Sylow p -normalizer). This numerical information alone may be sufficient to force $N_G(P) = G$ (i.e., $P \trianglelefteq G$), or at least to force $N_G(P)$ to have index smaller than the minimal index permitted by permutation representations, giving a contradiction by a preceding technique.

For example, there are no simple groups of order 1785. If there were, let G be a simple group of order $1785 = 3 \cdot 5 \cdot 7 \cdot 17$. The only possible value for n_{17} is 35, so if Q is a Sylow 17-subgroup, $|G : N_G(Q)| = 35$. Thus $|N_G(Q)| = 3 \cdot 17$. Let P be a Sylow 3-subgroup of $N_G(Q)$. The group PQ is abelian since 3 does not divide $17 - 1$, so $Q \leq N_G(P)$ and $17 \mid |N_G(P)|$. In this case $P \in \text{Syl}_3(G)$. The permissible values of n_3 are 7, 85 and 595; however, since $17 \mid |N_G(P)|$, we cannot have $17 \mid |G : N_G(P)| = n_3$. Thus $n_3 = 7$. But G has no proper subgroup of index < 17 (the minimal index of a proper subgroup is 17 for this order), a contradiction. Alternatively, if $n_3 = 7$, then $|N_G(P)| = 3 \cdot 5 \cdot 17$, and by Sylow's Theorem applied in $N_G(P)$ we have $Q \leq N_G(P)$. This contradicts the fact that $|N_G(Q)| = 3 \cdot 17$.

We can refine this method by not requiring P and Q to be of prime order. Namely, if p and q are distinct primes dividing $|G|$ such that $Q \in \text{Syl}_q(G)$ and $p \mid |N_G(Q)|$, let $P \in \text{Syl}_p(N_G(Q))$. We can then apply Sylow's Theorems in $N_G(Q)$ to see whether

$P \trianglelefteq N_G(Q)$, and if so, force $N_G(P)$ to be of small index. If P is a Sylow p -subgroup of the whole group G , we can use the congruence part of Sylow's Theorem to put further restrictions on $|N_G(P)|$ (as we did in the preceding example). If P is not a Sylow p -subgroup of G , then by the second part of Sylow's Theorem $P \leq P^* \in \text{Syl}_p(G)$. In this case since $P < P^*$, Theorem 1(4) shows that $P < N_{P^*}(P)$. Thus $N_G(P)$ (which contains $N_{P^*}(P)$) has order divisible by a larger power of p than divides $|P|$ (as well as being divisible by $|Q|$).

For example, there are no simple groups of order 3675. If there were, let G be a simple group of order $3675 = 3 \cdot 5^2 \cdot 7^2$. The only possibility for n_7 is 15, so for $Q \in \text{Syl}_7(G)$, $|G : N_G(Q)| = 15$ and $|N_G(Q)| = 245 = 5 \cdot 7^2$. Let $N = N_G(Q)$ and let $P \in \text{Syl}_5(N)$. By the congruence conditions of Sylow's Theorem applied in N we get $P \trianglelefteq N$. Since $|P| = 5$, P is not itself a Sylow 5-subgroup of G so P is contained in some Sylow 5-subgroup P^* of G . Since P is of index 5 in the 5-group P^* , $P \trianglelefteq P^*$ by Theorem 1, that is $P^* \leq N_G(P)$. This proves

$$\langle N, P^* \rangle \leq N_G(P) \quad \text{so} \quad 7^2 \cdot 5^2 \mid |N_G(P)|.$$

Thus $|G : N_G(P)| \mid 3$, which is impossible since P is not normal and G has no subgroup of index 3.

Studying Normalizers of Intersections of Sylow p -Subgroups

One of the reasons the counting arguments in the first method above do not immediately generalize to Sylow subgroups which are not of prime order is because if $P \in \text{Syl}_p(G)$ for some prime p and $|P| = p^a$, $a \geq 2$, then it need not be the case that distinct conjugates of P intersect in the identity subgroup. If distinct conjugates of P *do* intersect in the identity, we can again count to find that the number of elements of p -power order is $n_p(|P| - 1)$.

Suppose, however, there exists $R \in \text{Syl}_p(G)$ with $R \neq P$ and $P \cap R \neq 1$. Let $P_0 = P \cap R$. Then $P_0 < P$ and $P_0 < R$, hence by Theorem 1

$$P_0 < N_P(P_0) \quad \text{and} \quad P_0 < N_R(P_0).$$

One can try to use this to prove that the normalizer in G of P_0 is sufficiently large (i.e., of sufficiently small index) to obtain a contradiction by previous methods (note that this normalizer is a proper subgroup since $P_0 \neq 1$).

One special case where this works particularly well is when $|P_0| = p^{a-1}$ i.e., the two Sylow p -subgroups R and P have large intersection. In this case set $N = N_G(P_0)$. Then by the above reasoning (i.e., since P_0 is a maximal subgroup of the p -groups P and R), $P_0 \trianglelefteq P$ and $P_0 \trianglelefteq R$, that is,

N has 2 distinct Sylow p -subgroups: P and R .

In particular, $|N| = p^a k$, where (by Sylow's Theorem) $k \geq p + 1$.

Recapitulating, if Sylow p -subgroups pairwise intersect in the identity, then counting elements of p -power order is possible; otherwise there is some intersection of Sylow p -subgroups whose normalizer is "large." Since for an arbitrary group order one cannot necessarily tell which of these two phenomena occurs, it may be necessary to split the nonsimplicity argument into two (mutually exclusive) cases and derive a contradiction

in each. This process is especially amenable when the order of a Sylow p -subgroup is p^2 (for example, this line of reasoning was used to count elements of 2-power order in the proof that a simple group of order 60 is isomorphic to A_5 — Proposition 23, Section 4.5).

Before proceeding with an example we state a lemma which gives a sufficient condition to force a nontrivial Sylow intersection.

Lemma 13. In a finite group G if $n_p \not\equiv 1 \pmod{p^2}$, then there are distinct Sylow p -subgroups P and R of G such that $P \cap R$ is of index p in both P and R (hence is normal in each).

Proof: The argument is an easy refinement of the proof of the congruence part of Sylow's Theorem (cf. the exercises at the end of Section 4.5). Let P act by conjugation on the set $\text{Syl}_p(G)$. Let $\mathcal{O}_1, \dots, \mathcal{O}_s$ be the orbits under this action with $\mathcal{O}_1 = \{P\}$. If p^2 divides $|P : P \cap R|$ for all Sylow p -subgroups R of G different from P , then each $\mathcal{O}_i$ has size divisible by p^2 , $i = 2, 3, \dots, s$. In this case, since n_p is the sum of the lengths of the orbits we would have $n_p = 1 + kp^2$, contrary to assumption. Thus for some $R \in \text{Syl}_p(G)$, $|P : P \cap R| = p$.

For example, there are no simple groups of order 1053. If there were, let G be a simple group of order $1053 = 3^4 \cdot 13$ and let $P \in \text{Syl}_3(G)$. We must have $n_3 = 13$. But $13 \not\equiv 1 \pmod{3^2}$ so there exist $P, R \in \text{Syl}_3(G)$ such that $|P \cap R| = 3^3$. Let $N = N_G(P \cap R)$, so by the above arguments $P, R \leq N$. Thus $3^4 \mid |N|$ and $|N| > 3^4$. The only possibility is $N = G$, i.e., $P \cap R \leq G$, a contradiction.

Simple Groups of Order 168

We now show how many of our techniques can be used to unravel the structure of and then classify certain simple groups by classifying the simple groups of order 168. Because there are no nontrivial normal subgroups in simple groups, this process departs from the methods in Section 5.5, but the overall approach typifies methods used in the study of finite simple groups.

We begin by assuming there is a simple group G of order $168 = 2^3 \cdot 3 \cdot 7$. We first work out many of its properties: the number and structure of its Sylow subgroups, the conjugacy classes, etc. All of these calculations are based only on the order and simplicity of G . We use these results to first prove the uniqueness of G ; and ultimately we prove the existence of the simple group of order 168.

Because $|G|$ does not divide 6! we have

(1) G has no proper subgroup of index less than 7,

since otherwise the action of G on the cosets of the subgroup would give a (necessarily injective since G is simple) homomorphism from G into some S_n with $n \leq 6$.

The simplicity of G and Sylow's Theorem also immediately imply that

(2) $n_7 = 8$, so the normalizer of a Sylow 7-subgroup has order 21. In particular, no element of order 2 normalizes a Sylow 7-subgroup and G has no elements of order 14.

If G had an element of order 21 then the normalizer of a Sylow 3-subgroup of G would have order divisible by 7. Thus n_3 would be relatively prime to 7. Since then $n_3 \mid 8$ we would have $n_3 = 4$ contrary to (1). This proves:

(3) G has no elements of order 21.

By Sylow's Theorem $n_3 = 7$ or 28; we next rule out the former possibility. Assume $n_3 = 7$, let $P \in \text{Syl}_3(G)$ and let T be a Sylow 2-subgroup of the group $N_G(P)$ of order 24. Each Sylow 3-subgroup normalizes some Sylow 7-subgroup of G so P normalizes a Sylow 7-subgroup R of G . For every $t \in T$ we also have that $P = tPt^{-1}$ normalizes tRt^{-1} . The subgroup T acts by conjugation on the set of eight Sylow 7-subgroups of G and since no element of order 2 in G normalizes a Sylow 7-subgroup by (2), it follows that T acts transitively, i.e., every Sylow 7-subgroup of G is one of the tRt^{-1} . Hence P normalizes every Sylow 7-subgroup of G , i.e., P is contained in the intersection of the normalizers of all Sylow 7-subgroups. But this intersection is a proper normal subgroup of G , so it must be trivial. This contradiction proves:

(4) $n_3 = 28$ and the normalizer of a Sylow 3-subgroup has order 6.

Since $n_2 = 7$ or 21, we have $n_2 \not\equiv 1 \pmod{8}$, so by Exercise 21 there is a pair of distinct Sylow 2-subgroups that have nontrivial intersection; over all such pairs let T_1 and T_2 be chosen with $U = T_1 \cap T_2$ of maximal order. We next prove

(5) U is a Klein 4-group and $N_G(U) \cong S_4$.

Let $N = N_G(U)$. Since $|U| = 2$ or 4 and N permutes the nonidentity elements of U by conjugation, a subgroup of order 7 in N would commute with some element of order 2 in U , contradicting (2). It follows that the order of N is not divisible by 7. By Exercise 13, N has more than one Sylow 2-subgroup, hence $|N| = 2^a \cdot 3$, where $a = 2$ or 3. Let $P \in \text{Syl}_3(N)$. Since P is a Sylow 3-subgroup of G , by (4) the group $N_N(P)$ has order 3 or 6 (with P as its unique subgroup of order 3). Thus by Sylow's Theorem N must have four Sylow 3-subgroups, and these are permuted transitively by N under conjugation. Since any group of order 12 must have either a normal Sylow 2-subgroup or a normal Sylow 3-subgroup (cf. Section 4.5), $|N| = 24$. Let K be the kernel of N acting by conjugation on its four Sylow 3-subgroups, so K is the intersection of the normalizers of the Sylow 3-subgroups of N . If $K = 1$ then $N \cong S_4$ as asserted; so consider when $K \neq 1$. Since $K \leq N_N(P)$, the group K has order dividing 6, and since P does not normalize another Sylow 3-subgroup, P is not contained in K . It follows that $|K| = 2$. But now N/K is a group of order 12 which is seen to have more than one Sylow 2-subgroup and four Sylow 3-subgroups, contrary to the property of groups of order 12 cited earlier. This proves $N \cong S_4$. Since S_4 has a unique nontrivial normal 2-subgroup, V_4 , (5) holds. Since $N \cong S_4$, it follows that N contains a Sylow 2-subgroup of G and also that $N_N(P) \cong S_3$ (so also $N_G(P) \cong S_3$ by (4)). Hence we obtain

(6) Sylow 2-subgroups of G are isomorphic to D_8 , and

(7) the normalizer in G of a Sylow 3-subgroup is isomorphic to S_3 and so G has no elements of order 6.

By (2) and (7), no element of order 2 commutes with an element of odd prime order. If $T \in \text{Syl}_2(G)$, then $T \cong D_8$ by (6), so $Z(T) = \langle z \rangle$ where z is an element of order 2. Then $T \leq C_G(z)$ and $|C_G(z)|$ has no odd prime factors by what was just said, so $C_G(z) = T$. Since any element normalizing T would normalize its center, hence commute with z , it follows that Sylow 2-subgroups of G are self-normalizing. This gives

(8) $n_2 = 21$ and $C_G(z) = T$, where $T \in \text{Syl}_2(G)$ and $Z(T) = \langle z \rangle$.

Since $|C_G(z)| = 8$, the element z in (8) has 21 conjugates. By (6), G has one conjugacy class of elements of order 4, which by (6) and (8) contains 42 elements. By (2) there are 48 elements of order 7, and by (4) there are 56 elements of order 3. These account for all 167 nonidentity elements of G , and so every element of order 2 must be conjugate to z , i.e.,

(9) G has a unique conjugacy class of elements of order 2.

Continuing with the same notation, let $T \in \text{Syl}_2(G)$ with $U \leq T$ and let W be the other Klein 4-group in T . It follows from Sylow's Theorem that U and W are not conjugate in G since they are not conjugate in $N_G(T) = T$ (cf. Exercise 50 in Section 4.5). We argue next that

(10) $N_G(W) \cong S_4$.

To see this let $W = \langle z, w \rangle$ where, as before, $\langle z \rangle = Z(T)$. Since w is conjugate in G to z , $C_G(w) = T_0$ is another Sylow 2-subgroup of G containing W but different from T . Thus $W = T \cap T_0$. Since U was an arbitrary maximal intersection of Sylow 2-subgroups of G , the argument giving (5) implies (10).

We now record results which we have proved or which are easy consequences of (1) to (10).

Proposition 14. If G is a simple group of order 168, then the following hold:

- (1) $n_2 = 21$, $n_3 = 7$ and $n_7 = 8$
- (2) Sylow 2-subgroups of G are dihedral, Sylow 3- and 7-subgroups are cyclic
- (3) G is isomorphic to a subgroup of A_7 and G has no subgroup of index ≤ 6
- (4) the conjugacy classes of G are the following: the identity; two classes of elements of order 7 each of which contains 24 elements (represented by any element of order 7 and its inverse); one class of elements of order 3 containing 56 elements; one class of elements of order 4 containing 42 elements; one class of elements of order 2 containing 21 elements
(in particular, every element of G has order a power of a prime)
- (5) if $T \in \text{Syl}_2(G)$ and U, W are the two Klein 4-groups in T , then U and W are not conjugate in G and $N_G(U) \cong N_G(W) \cong S_4$
- (6) G has precisely three conjugacy classes of maximal subgroups, two of which are isomorphic to S_4 and one of which is isomorphic to the non-abelian group of order 21.

All of the calculations above were predicated on the assumption that there exists a simple group of order 168. The fact that none of these arguments leads to a contradiction

does not *prove* the existence of such a group, but rather just gives strong evidence that there *may* be a simple group of this order. We next illustrate how the internal subgroup structure of G gives rise to a geometry on which G acts, and so leads to a proof that a simple group of order 168 is unique, if it exists (which we shall also show).

Continuing the above notation let $U_1, \dots, U_7$ be the conjugates of U and let $W_1, \dots, W_7$ be the conjugates of W . Call the U_i *points* and the W_j *lines*. Define an “incidence relation” by specifying that

the point U_i is on the line W_j if and only if U_i normalizes W_j .

Note that U_i normalizes W_j if and only if $U_i W_j \cong D_8$, which in turn occurs if and only if W_j normalizes U_i . In each point or line stabilizer—which is isomorphic to S_4 —there is a unique normal 4-group, V , and precisely three other (nonnormal) 4-groups A_1, A_2, A_3 . The groups VA_i are the three Sylow 2-subgroups of the S_4 . We therefore have:

(11) *each line contains exactly 3 points and each point lies on exactly 3 lines.*

Since any two nonnormal 4-groups in an S_4 generate the S_4 , hence uniquely determine the other two Klein groups in that S_4 , we obtain

(12) *any 2 points on a line uniquely determine the line (and the third point on it).*

Since there are 7 points and 7 lines, elementary counting now shows that

(13) *each pair of points lies on a unique line, and each pair of lines intersects in a unique point.*

(This configuration of points and lines thus satisfies axioms for what is termed a *projective plane*.) It is now straightforward to show that the incidence geometry is uniquely determined and may be represented by the graph in Figure 1, where points are vertices and lines are the six sides and medians of the triangle together with the inscribed circle—see Exercise 27. This incidence geometry is called the *projective plane of order 2* or the *Fano Plane*, and will be denoted by $\mathcal{F}$. (Generally, a projective plane of “order” N has $N^2 + N + 1$ points, and the same number of lines.) Note that at this point the projective plane $\mathcal{F}$ *does* exist—we have explicitly exhibited points and lines satisfying (11) to (13)—even though the group G is not yet known to exist.

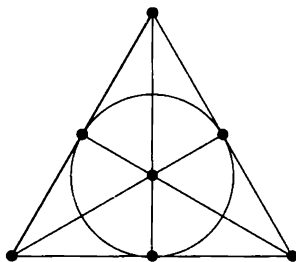


Figure 1

An *automorphism* of this plane is any permutation of points and lines that preserves the incidence relation. For example, any of the six symmetries of the triangle in Figure 1

give automorphisms of $\mathcal{F}$, but we shall see that $\mathcal{F}$ has many more automorphisms than these.

Each $g \in G$ acts by conjugation on the set of points and lines, and this action preserves the incidence relation. Only the identity element in G fixes all points and so via this action the group G would be isomorphic to a subgroup of the group of $\text{Aut}(\mathcal{F})$, the group of all automorphisms of $\mathcal{F}$.

Any automorphism of $\mathcal{F}$ that fixes two points on a line as well as a third point not on that line is easily seen to fix all points. Thus any automorphism of $\mathcal{F}$ is uniquely determined by its action on any three noncollinear points. Since one easily computes that there are 168 such triples, $\mathcal{F}$ has at most 168 automorphisms. This proves

if the simple group G exists it is unique and $G \cong \text{Aut}(\mathcal{F})$.

Two steps in the classification process yet remain: to prove that $\mathcal{F}$ does have 168 automorphisms and to prove $\text{Aut}(\mathcal{F})$ is indeed a simple group. Although one can do these graph-theoretically, we adopt an approach following ideas from the theory of “algebraic groups.” Let V be a 3-dimensional vector space over the field of 2 elements, $\mathbb{F}_2$, so V is the elementary abelian 2-group $Z_2 \times Z_2 \times Z_2$ of order 8. By Proposition 17 in Section 4.4, $\text{Aut}(V) = GL(V) \cong GL_3(\mathbb{F}_2)$ has order 168. Call the seven 1-dimensional subspaces (i.e., the nontrivial cyclic subgroups) of V *points*, call the seven 2-dimensional subspaces (i.e., the subgroups of order 4) *lines*, and say the point p is *incident to* the line L if $p \subset L$. Then the points and lines are easily seen to satisfy the same axioms (11) to (13) above, hence to represent the Fano Plane. Since $GL(V)$ acts faithfully on these points and lines preserving incidence, $\text{Aut}(\mathcal{F})$ has order at least 168. In light of the established upper bound for $|\text{Aut}(\mathcal{F})|$ this proves

$\text{Aut}(\mathcal{F}) \cong GL(V) \cong GL_3(\mathbb{F}_2)$ and $\text{Aut}(\mathcal{F})$ has order 168.

Finally we prove that $GL(V)$ is a simple group. By way of contradiction assume H is a proper nontrivial normal subgroup of $GL(V)$. Let Ω be the 7 points and let N be the stabilizer in $GL(V)$ of some point in Ω . Since $GL(V)$ acts transitively on Ω , N has index 7. Since the intersection of all conjugates of N fixes all points, this intersection is the identity. Thus $H \not\leq N$, and so $GL(V) = HN$. Since $|H : H \cap N| = |HN : N|$ we have $7 \mid |H|$. Since $GL(V)$ is isomorphic to a subgroup of S_7 and since Sylow 7-subgroups of S_7 have normalizers of order 42, $GL(V)$ does not have a normal Sylow 7-subgroup, so by Sylow’s Theorem $n_7(GL(V)) = 8$. A normal Sylow 7-subgroup of H would be characteristic in H , hence normal in $GL(V)$, so also H does not have a unique Sylow 7-subgroup. Since $n_7(H) \equiv 1 \pmod{7}$ and $n_7(H) \leq n_7(GL(V)) = 8$ we must have $n_7(H) = 8$. This implies $|H|$ is divisible by 8, so $56 \mid |H|$, and since H is proper we must have $|H| = 56$. By usual counting arguments (cf. Exercise 7(b) of Section 5.5) H has a normal, hence characteristic, Sylow 2-subgroup, which is therefore normal in $GL(V)$. But then $GL(V)$ would have a unique Sylow 2-subgroup. Since the set of upper triangular matrices and the set of lower triangular matrices are two subgroups of $GL_3(\mathbb{F}_2)$ each of order 8, we have a contradiction. In summary we have now proven the following theorem.

Theorem 15. Up to isomorphism there is a unique simple group of order 168, $GL_3(\mathbb{F}_2)$, which is also the automorphism group of the projective plane $\mathcal{F}$.

Note that we might just as well have called the W_j points and the U_i lines. This “duality” between points and lines together with the uniqueness of a simple group of order 168 may be used to prove the existence of an outer automorphism of G that interchanges points and lines i.e., conjugates U to W .

Many families of finite simple groups can be classified by analogous methods. In more general settings geometric structures known as *buildings* play the role of the projective plane (which is a special case of a building of type A_2). In this context the subgroups $N_G(U)$ and $N_G(W)$ are *parabolic subgroups* of G , and U , W are their *unipotent radicals* respectively. In particular, all the simple linear groups (cf. Section 3.4) are characterized by the structure and intersections of their parabolic subgroups, or equivalently, by their action on an associated building.

Remarks on the Existence Problem for Groups

As in other areas of mathematics (such as the theory of differential equations) one may hypothesize the existence of a mathematical system (e.g., solution to an equation) and derive a great deal of information about this proposed system. In general, if after considerable effort no contradiction is reached based on the initial hypothesis one begins to suspect that there actually is a system which does satisfy the conditions hypothesized. However, no amount of consistent data will *prove* existence. Suppose we carried out an analysis of a hypothetical simple group G of order $3^3 \cdot 7 \cdot 13 \cdot 409$ analogous to our analysis of a simple group of order 168 (which we showed to exist). After a certain amount of effort we could show that there are unique possible Sylow numbers:

$$n_3 = 7 \cdot 409 \quad n_7 = 3^2 \cdot 13 \cdot 409 \quad n_{13} = 3^2 \cdot 7 \cdot 409 \quad n_{409} = 3^2 \cdot 7 \cdot 13.$$

We could further show that such a G would have no elements of order pq , p and q distinct primes, no elements of order 9, and that distinct Sylow subgroups would intersect in the identity. We could then count the elements in Sylow p -subgroups for all primes p and we would find that these would total to exactly $|G|$. At this point we would have the complete subgroup structure and class equation for G . We might then guess that there is a simple group of this order, but the Feit–Thompson Theorem asserts that there are *no* simple groups of odd composite order. (Note, however, that the configuration for a possible simple group of order $3^3 \cdot 7 \cdot 13 \cdot 409$ is among the cases that must be dealt with in the *proof* of the Feit–Thompson Theorem, so quoting this result in this instance is actually circular. We prove no simple group of this order exists in Section 19.3; see also Exercise 29.) The point is that even though we have as much data in this case as we had in the order 168 situation (i.e., Proposition 14), we cannot prove existence without some new techniques.

When we are dealing with nonsimple groups we have at least one method of building larger groups from smaller ones: semidirect products. Even though this method is fairly restrictive it conveys the notion that nonsimple groups may be built up from smaller groups in some constructive fashion. This process breaks down completely for simple groups; and so this demarcation of techniques reinforces our appreciation for the Hölder

Program: determining the simple groups, and finding how these groups are put together to form larger groups.

The study of simple groups, as illustrated in the preceding discussion of groups of order 168, uses many of the same tools as the study of nonsimple groups (to unravel their subgroup structures, etc.) but also requires other techniques for their construction. As we mentioned at the end of that discussion, these often involve algebraic or geometric methods which construct simple groups as automorphisms of mathematical structures that have intrinsic interest, and thereby link group theory to other areas of mathematics and science in fascinating ways. Thus while we have come a long way in the analysis of finite groups, there are a number of different areas in this branch of mathematics on which we have just touched.

The analysis of infinite groups generally involves quite different methods, and in the next section we introduce some of these.

EXERCISES

Counting elements:

1. Prove that for fixed $P \in \text{Syl}_p(G)$ if $P \cap R = 1$ for all $R \in \text{Syl}_p(G) - \{P\}$, then $P_1 \cap P_2 = 1$ whenever P_1 and P_2 are distinct Sylow p -subgroups of G . Deduce in this case that the number of nonidentity elements of p -power order in G is $(|P| - 1)|G : N_G(P)|$.
2. In the group $S_3 \times S_3$ exhibit a pair of Sylow 2-subgroups that intersect in the identity and exhibit another pair that intersect in a group of order 2.
3. Prove that if $|G| = 380$ then G is not simple. [Just count elements of odd prime order.]
4. Prove that there are no simple groups of order 80, 351, 3875 or 5313.
5. Let G be a solvable group of order pm , where p is a prime not dividing m , and let $P \in \text{Syl}_p(G)$. If $N_G(P) = P$, prove that G has a normal subgroup of order m . Where was the solvability of G needed in the proof? (This result is true for nonsolvable groups as well — it is a special case of *Burnside's N/C-Theorem*.)

Exploiting subgroups of small index:

6. Prove that there are no simple groups of order 2205, 4125, 5103, 6545 or 6435.

Permutation representations:

7. Prove that there are no simple groups of order 1755 or 5265. [Use Sylow 3-subgroups to show $G \leq S_{13}$ and look at the normalizer of a Sylow 13-subgroup.]
8. Prove that there are no simple groups of order 792 or 918.
9. Prove that there are no simple groups of order 336.

Playing p -subgroups off against each other:

10. Prove that there are no simple groups of order 4095, 4389, 5313 or 6669.
11. Prove that there are no simple groups of order 4851 or 5145.
12. Prove that there are no simple groups of order 9555. [Let $Q \in \text{Syl}_{13}(G)$ and let $P \in \text{Syl}_7(N_G(Q))$. Argue that $Q \trianglelefteq N_G(P)$ — why is this a contradiction?]

Normalizers of Sylow intersections:

13. Let G be a group with more than one Sylow p -subgroup. Over all pairs of distinct Sylow p -subgroups let P and Q be chosen so that $|P \cap Q|$ is maximal. Show that $N_G(P \cap Q)$

has more than one Sylow p -subgroup and that any two distinct Sylow p -subgroups of $N_G(P \cap Q)$ intersect in the subgroup $P \cap Q$. (Thus $|N_G(P \cap Q)|$ is divisible by $p \cdot |P \cap Q|$ and by some prime other than p . Note that Sylow p -subgroups of $N_G(P \cap Q)$ need not be Sylow in G .)

14. Prove that there are no simple groups of order 144, 525, 2025 or 3159.

General exercises:

15. Classify groups of order 105.
16. Prove that there are no non-abelian simple groups of odd order < 10000 .
17. (a) Prove that there is no simple group of order 420.
(b) Prove that there are no simple groups of even order < 500 except for orders 2, 60, 168 and 360.
18. Prove that if G is a group of order 36 then G has either a normal Sylow 2-subgroup or a normal Sylow 3-subgroup.
19. Show that a group of order 12 with no subgroup of order 6 is isomorphic to A_4 .
20. Show that a group of order 24 with no element of order 6 is isomorphic to S_4 .
21. Generalize Lemma 13 by proving that if $n_p \not\equiv 1 \pmod{p^k}$ then there are distinct Sylow p -subgroups P and R of G such that $P \cap R$ is of index $\leq p^{k-1}$ in both P and R .
22. Suppose over all pairs of distinct Sylow p -subgroups of G , P and R are chosen with $|P \cap R|$ maximal. Prove that $N_G(P \cap R)$ is not a p -group.
23. Let A and B be normal subsets of a Sylow p -subgroup P of G . Prove that if A and B are conjugate in G then they are conjugate in $N_G(P)$.
24. Let G be a group of order pqr where p, q and r are primes with $p < q < r$. Prove that a Sylow r -subgroup of G is normal.
25. Let G be a simple group of order p^2qr where p, q and r are primes. Prove that $|G| = 60$.
26. Prove or construct a counterexample to the assertion: if G is a group of order 168 with more than one Sylow 7-subgroup then G is simple.
27. Show that if $\mathcal{F}$ is any set of points and lines satisfying properties (11) to (13) in the subsection on simple groups of order 168 then the graph of incidences for $\mathcal{F}$ is uniquely determined and is the same as Figure 1 (up to relabeling points and lines). [Take a line and any point not on this line. Depict the line as the base of an equilateral triangle and the point as the vertex of this triangle not on the base. Use the axioms to show that the incidences of the remaining points and lines are then uniquely determined as in Figure 1.]
28. Let G be a simple group of order $3^3 \cdot 7 \cdot 13 \cdot 409$. Compute all permissible values of n_p for each $p \in \{3, 7, 13, 409\}$ and reduce to the case where there is a unique possible value for each n_p .
29. Given the information on the Sylow numbers for a hypothetical simple group of order $3^3 \cdot 7 \cdot 13 \cdot 409$, prove that there is no such group. [Work with the permutation representation of degree 819.]
30. Suppose G is a simple group of order 720. Find as many properties of G as you can (Sylow numbers, isomorphism type of Sylow subgroups, conjugacy classes, etc.). Is there such a group?

6.3 A WORD ON FREE GROUPS

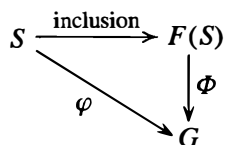
In this section we introduce the basic theory of so-called free groups. This will enable us to make precise the notions of generators and relations which were used in earlier chapters. The results of this section rely only on the basic theory of homomorphisms.

The basic idea of a free group $F(S)$ generated by a set S is that there are no relations satisfied by any of the elements in S (S is “free” of relations). For example, if S is the set $\{a, b\}$ then the elements of the free group on the two generators a and b are of the form $a, aa, ab, abab, bab$, etc., called *words* in a and b , together with the inverses of these elements, and all these elements are considered distinct. If we group like terms together, then we obtain elements of the familiar form $a, b^{-3}, aba^{-1}b^2$ etc. Such elements are multiplied by concatenating their words (for example, the product of aba and $b^{-1}a^3b$ would simply be $abab^{-1}a^3b$). It is natural at the outset (even before we know S is contained in some group) to simply *define* $F(S)$ to be the set of all words in S , where two such expressions are multiplied in $F(S)$ by concatenating them. Although in essence this is what we do, it is necessary to be more formal in order to prove that this concatenation operation is well defined and associative. After all, even the familiar notation a^n for the product $a \cdot a \cdots a$ (n terms) is permissible only because we know that this product is independent of the way it is bracketed (cf. the generalized associative law in Section 1.1). The formal construction of $F(S)$ is carried out below for an arbitrary set S .

One important property reflecting the fact that there are no relations that must be satisfied by the generators in S is that any *map* from the *set* S to a group G can be uniquely extended to a *homomorphism* from the *group* $F(S)$ to G (basically since we have specified where the generators must go and the images of all the other elements are uniquely determined by the homomorphism property — the fact that there are no relations to worry about means that we can specify the images of the generators *arbitrarily*). This is frequently referred to as the *universal* property of the free group and in fact characterizes the group $F(S)$.

The notion of “freeness” occurs in many algebraic systems and it may already be familiar (using a different terminology) from elementary vector space theory. When the algebraic systems are vector spaces, $F(S)$ is simply the vector space which has S as a basis. Every vector in this space is a unique linear combination of the elements of S (the analogue of a “word”). Any set map from the basis S to another vector space V extends uniquely to a linear transformation (i.e., vector space homomorphism) from $F(S)$ to V .

Before beginning the construction of $F(S)$ we mention that one often sees the universal property described in the language of commutative diagrams. In this form it reads (for groups) as follows: given any set map φ from the set S to a group G there is a unique homomorphism $\Phi : F(S) \rightarrow G$ such that $\Phi|_S = \varphi$ i.e., such that the following diagram commutes:



As mentioned above, the only difficulty with the construction of $F(S)$ is the verification that the concatenation operation on the words in $F(S)$ is well defined and associative. To prove the associative property for multiplication of words we return to the most basic level where all the exponents in the words of S are ± 1 .

We first introduce inverses for elements of S and an identity.

Let S^{-1} be any set disjoint from S such that there is a bijection from S to S^{-1} . For each $s \in S$ denote its corresponding element in S^{-1} by s^{-1} and similarly for each $t \in S^{-1}$ let the corresponding element of S be denoted by t^{-1} (so $(s^{-1})^{-1} = s$). Take a singleton set not contained in $S \cup S^{-1}$ and call it $\{1\}$. Let $1^{-1} = 1$ and for any $x \in S \cup S^{-1} \cup \{1\}$ let $x^{-1} = x$.

Next we describe the elements of the free group on the set S . A *word* on S is by definition a sequence

$$(s_1, s_2, s_3, \dots) \quad \text{where } s_i \in S \cup S^{-1} \cup \{1\} \text{ and } s_i = 1 \text{ for all } i \text{ sufficiently large}$$

(that is, for each sequence there is an N such that $s_i = 1$ for all $i \geq N$). Thus we can think of a word as a finite product of elements of S and their inverses (where repetitions are allowed). Next, in order to assure uniqueness of expressions we consider only words which have no obvious "cancellations" between adjacent terms (such as $baa^{-1}b = bb$). The word $(s_1, s_2, s_3, \dots)$ is said to be *reduced* if

- (1) $s_{i+1} \neq s_i^{-1}$ for all i with $s_i \neq 1$, and
- (2) if $s_k = 1$ for some k , then $s_i = 1$ for all $i \geq k$.

The reduced word $(1, 1, 1, \dots)$ is called the *empty word* and is denoted by 1. We now simplify the notation by writing the reduced word $(s_1^{\epsilon_1}, s_2^{\epsilon_2}, \dots, s_n^{\epsilon_n}, 1, 1, 1, \dots)$, $s_i \in S$, $\epsilon_i = \pm 1$, as $s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}$. Note that by definition, reduced words $r_1^{\delta_1} r_2^{\delta_2} \dots r_m^{\delta_m}$ and $s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}$ are equal if and only if $n = m$ and $\delta_i = \epsilon_i$, $1 \leq i \leq n$. Let $F(S)$ be the set of reduced words on S and embed S into $F(S)$ by

$$s \mapsto (s, 1, 1, 1, \dots).$$

Under this set injection we identify S with its image and henceforth consider S as a subset of $F(S)$. Note that if $S = \emptyset$, $F(S) = \{1\}$.

We are now in a position to introduce the binary operation on $F(S)$. The principal technical difficulty is to ensure that the product of two reduced words is again a *reduced* word. Although the definition appears to be complicated it is simply the formal rule for "successive cancellation" of juxtaposed terms which are inverses of each other (e.g., $ab^{-1}a$ times $a^{-1}ba$ should reduce to aa). Let $r_1^{\delta_1} r_2^{\delta_2} \dots r_m^{\delta_m}$ and $s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}$ be reduced words and assume first that $m \leq n$. Let k be the smallest integer in the range $1 \leq k \leq m+1$ such that $s_k^{\epsilon_k} \neq r_{m-k+1}^{-\delta_{m-k+1}}$. Then the product of these reduced words is defined to be:

$$(r_1^{\delta_1} r_2^{\delta_2} \dots r_m^{\delta_m})(s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}) = \begin{cases} r_1^{\delta_1} \dots r_{m-k+1}^{\delta_{m-k+1}} s_k^{\epsilon_k} \dots s_n^{\epsilon_n}, & \text{if } k \leq m \\ s_{m+1}^{\epsilon_{m+1}} \dots s_n^{\epsilon_n}, & \text{if } k = m+1 \leq n \\ 1, & \text{if } k = m+1 \text{ and } m = n. \end{cases}$$

The product is defined similarly when $m \geq n$, so in either case it results in a reduced word.

Theorem 16. $F(S)$ is a group under the binary operation defined above.

Proof: One easily checks that 1 is an identity and that the inverse of the reduced word $s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}$ is the reduced word $s_n^{-\epsilon_n} s_{n-1}^{-\epsilon_{n-1}} \dots s_1^{-\epsilon_1}$. The difficult part of the proof is the verification of the associative law. This can be done by induction on the “length” of the words involved and considering various cases or one can proceed as follows: For each $s \in S \cup S^{-1} \cup \{1\}$ define $\sigma_s : F(S) \rightarrow F(S)$ by

$$\sigma_s(s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}) = \begin{cases} s \cdot s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}, & \text{if } s_1^{\epsilon_1} \neq s^{-1} \\ s_2^{\epsilon_2} s_3^{\epsilon_3} \dots s_n^{\epsilon_n}, & \text{if } s_1^{\epsilon_1} = s^{-1}. \end{cases}$$

Since $\sigma_{s^{-1}} \circ \sigma_s$ is the identity map of $F(S) \rightarrow F(S)$, σ_s is a permutation of $F(S)$. Let $A(F)$ be the subgroup of the symmetric group on the set $F(S)$ which is generated by $\{\sigma_s \mid s \in S\}$. It is easy to see that the map

$$s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n} \mapsto \sigma_{s_1}^{\epsilon_1} \circ \sigma_{s_2}^{\epsilon_2} \circ \dots \circ \sigma_{s_n}^{\epsilon_n}$$

is a (set) bijection between $F(S)$ and $A(S)$ which respects their binary operations. Since $A(S)$ is a group, hence associative, so is $F(S)$.

The universal property of free groups now follows easily.

Theorem 17. Let G be a group, S a set and $\varphi : S \rightarrow G$ a set map. Then there is a unique group homomorphism $\Phi : F(S) \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} S & \xrightarrow{\text{inclusion}} & F(S) \\ & \searrow \varphi & \downarrow \Phi \\ & & G \end{array}$$

Proof: Such a map Φ must satisfy $\Phi(s_1^{\epsilon_1} s_2^{\epsilon_2} \dots s_n^{\epsilon_n}) = \varphi(s_1)^{\epsilon_1} \varphi(s_2)^{\epsilon_2} \dots \varphi(s_n)^{\epsilon_n}$ if it is to be a homomorphism (which proves uniqueness), and it is straightforward to check that this map is in fact a homomorphism (which proves existence).

Corollary 18. $F(S)$ is unique up to a unique isomorphism which is the identity map on the set S .

Proof: This follows from the universal property. Suppose $F(S)$ and $F'(S)$ are two free groups generated by S . Since S is contained in both $F(S)$ and $F'(S)$, we have natural injections $S \hookrightarrow F'(S)$ and $S \hookrightarrow F(S)$. By the universal property in the theorem, it follows that we have unique associated group homomorphisms $\Phi : F(S) \rightarrow F'(S)$ and $\Phi' : F'(S) \rightarrow F(S)$ which are both the identity on S . The composite $\Phi' \Phi$ is a homomorphism from $F(S)$ to $F(S)$ which is the identity on S , so by the uniqueness statement in the theorem, it must be the identity map. Similarly $\Phi \Phi'$ is the identity, so Φ is an isomorphism (with inverse Φ'), which proves the corollary.

Definition. The group $F(S)$ is called the *free group* on the set S . A group F is a *free group* if there is some set S such that $F = F(S)$ — in this case we call S a set of *free generators* (or a *free basis*) of F . The cardinality of S is called the *rank* of the free group.

One can now simplify expressions in a free group by using exponential notation, so we write a^3b^{-2} instead of the formal reduced word $aaab^{-1}b^{-1}$. Expressions like aba , however, cannot be simplified in the free group on $\{a, b\}$. We mention one important theorem in this area.

Theorem 19. (Schreier) Subgroups of a free group are free.

This is not trivial to prove and we do not include a proof. There is a nice proof of this result using covering spaces (cf. *Trees* by J.-P. Serre, Springer-Verlag, 1980).

Presentations

Let G be any group. Then G is a homomorphic image of a free group: take $S = G$ and φ as the identity map from G to G ; then Theorem 16 produces a (surjective) homomorphism from $F(G)$ onto G . More generally, if S is any subset of G such that $G = \langle S \rangle$, then again there is a unique surjective homomorphism from $F(S)$ onto G which is the identity on S . (Note that we can now independently formulate the notion that a subset *generates* a group by noting that $G = \langle S \rangle$ if and only if the map $\pi : F(S) \rightarrow G$ which extends the identity map of S to G is surjective.)

Definition. Let S be a subset of a group G such that $G = \langle S \rangle$.

- (1) A *presentation* for G is a pair (S, R) , where R is a set of words in $F(S)$ such that the normal closure of $\langle R \rangle$ in $F(S)$ (the smallest normal subgroup containing $\langle R \rangle$) equals the kernel of the homomorphism $\pi : F(S) \rightarrow G$ (where π extends the identity map from S to G). The elements of S are called *generators* and those of R are called *relations* of G .
- (2) We say G is *finitely generated* if there is a presentation (S, R) such that S is a finite set and we say G is *finitely presented* if there is a presentation (S, R) with both S and R finite sets.

Note that if (S, R) is a presentation, the kernel of the map $F(S) \rightarrow G$ is not $\langle R \rangle$ itself but rather the (much larger) group generated by R and *all conjugates* of elements in R . Note that even for a fixed set S a group will have many different presentations (we can always throw redundant relations into R , for example). If G is finitely presented with $S = \{s_1, s_2, \dots, s_n\}$ and $R = \{w_1, w_2, \dots, w_k\}$, we write (as we have in preceding chapters):

$$G = \langle s_1, s_2, \dots, s_n \mid w_1 = w_2 = \dots = w_k = 1 \rangle$$

and if w is the word $w_1w_2^{-1}$, we shall write $w_1 = w_2$ instead of $w = 1$.

Examples

- (1) Every finite group is finitely presented. To see this let $G = \{g_1, \dots, g_n\}$ be a finite group. Let $S = G$ and let $\pi : F(S) \rightarrow G$ be the homomorphism extending the identity map of S . Let R_0 be the set of words $g_i g_j g_k^{-1}$, where $i, j = 1, \dots, n$ and $g_i g_j = g_k$ in G . Clearly $R_0 \leq \ker \pi$. If N is the normal closure of R_0 in $F(S)$ and $\tilde{G} = F(S)/N$, then G is a homomorphic image of $\tilde{G}$ (i.e., π factors through N). Moreover, the set of elements $\{\tilde{g}_i \mid i = 1, \dots, n\}$ is closed under multiplication. Since this set generates $\tilde{G}$, it must equal $\tilde{G}$. Thus $|\tilde{G}| = |G|$ and so $N = \ker \pi$ and (S, R_0) is a presentation of G .

This illustrates a sufficient condition for (S, R) to be a presentation for a given finite group G :

- (i) S must be a generating set for G , and
 - (ii) any group generated by S satisfying the relations in R must have order $\leq |G|$.
- (2) Abelian groups can be presented easily. For instance

$$\begin{aligned}\mathbb{Z} &\cong F(\{a\}) = \langle a \rangle, \\ \mathbb{Z} \times \mathbb{Z} &\cong \langle a, b \mid [a, b] = 1 \rangle, \\ \mathbb{Z}_n \times \mathbb{Z}_m &\cong \langle a, b \mid a^n = b^m = [a, b] = 1 \rangle.\end{aligned}$$

(Recall $[a, b] = a^{-1}b^{-1}ab$).

- (3) Some familiar non-abelian groups introduced in earlier chapters have simple presentations:

$$\begin{aligned}D_{2n} &= \langle r, s \mid r^n = s^2 = 1, s^{-1}rs = r^{-1} \rangle \\ Q_8 &= \langle i, j \mid i^4 = 1, j^2 = i^2, j^{-1}ij = i^{-1} \rangle.\end{aligned}$$

To check, for example, the presentation for D_{2n} note that the relations in the presentation $\langle r, s \mid r^n = s^2 = 1, s^{-1}rs = r^{-1} \rangle$ imply that this group has a normal subgroup (generated by r) of order $\leq n$ whose quotient is generated by s (which has order ≤ 2). Thus any group with these generators and relations has order at most $2n$. Since we already know of the existence of the group D_{2n} of order $2n$ satisfying these conditions, the abstract presentation must equal D_{2n} .

- (4) As mentioned in Section 1.2, in general it is extremely difficult even to determine if a given set of generators and relations is or is not the identity group (let alone determine whether it is some other nontrivial finite group). For example, in the following two presentations the first group is an *infinite* group and the second is the *identity* group (cf. *Trees*, Chapter 1):

$$\begin{aligned}\langle x_1, x_2, x_3, x_4 \mid x_2 x_1 x_2^{-1} &= x_1^2, x_3 x_2 x_3^{-1} = x_2^2, x_4 x_3 x_4^{-1} = x_3^2, x_1 x_4 x_1^{-1} = x_4^2 \rangle \\ \langle x_1, x_2, x_3, \mid x_2 x_1 x_2^{-1} &= x_1^2, x_3 x_2 x_3^{-1} = x_2^2, x_1 x_3 x_1^{-1} = x_3^2 \rangle.\end{aligned}$$

- (5) It is easy to see that S_n is generated by the transpositions $(1\ 2), (2\ 3), \dots, (n-1\ n)$, and that these satisfy the relations

$$((i\ i+1)(i+1\ i+2))^3 = 1 \quad \text{and} \quad [(i\ i+1), (j\ j+1)] = 1, \text{ whenever } |i - j| \geq 2$$

(here $|i - j|$ denotes the absolute value of the integer $i - j$). One can prove by induction on n that these form a presentation of S_n :

$$\begin{aligned}S_n &\cong \langle t_1, \dots, t_{n-1} \mid t_i^2 = 1, (t_i t_{i+1})^3 = 1, \text{ and } [t_i, t_j] = 1 \\ &\quad \text{whenever } |i - j| \geq 2, 1 \leq i, j \leq n-1 \rangle.\end{aligned}$$

As mentioned in Section 1.6 we can use presentations of a group to find homomorphisms between groups or to find automorphisms of a group. We did this in classifying groups of order 6, for example, when we proved that any non-abelian group of order 6 was generated by an element of order 3 and an element of order 2 inverting it; thus there is a homomorphism from S_3 onto any non-abelian group of order 6 (hence an isomorphism, by computing orders). More generally, suppose G is presented by, say, generators a, b with relations $r_1, \dots, r_k$. If a', b' are any elements of a group H satisfying these relations, there is a homomorphism from G into H . Namely, if $\pi : F(\{a, b\}) \rightarrow G$ is the presentation homomorphism, we can define $\pi' : F(\{a, b\}) \rightarrow H$ by $\pi'(a) = a'$ and $\pi'(b) = b'$. Then $\ker \pi \leq \ker \pi'$ so π' factors through $\ker \pi$ and we obtain

$$G \cong F(\{a, b\}) / \ker \pi \longrightarrow H.$$

In particular, if $\langle a', b' \rangle = H = G$, this homomorphism is an automorphism of G . Conversely, any automorphism must send a set of generators to another set of generators satisfying the same relations. For example, $D_8 = \langle a, b \mid a^2 = b^4 = 1, aba = b^{-1} \rangle$ and any pair a', b' of elements, where a' is a noncentral element of order 2 and b' is of order 4, satisfies the same relations. Since there are four noncentral elements of order 2 and two elements of order 4, D_8 has 8 automorphisms.

Similarly, any pair of elements of order 4 in Q_8 which are not equal or inverses of each other necessarily generate Q_8 and satisfy the relations given in Example 3 above. It is easy to check that there are 24 such pairs, so

$$|\text{Aut}(Q_8)| = 24.$$

Free objects can be constructed in (many, but not all) other categories. For instance, a *monoid* is a set together with a binary operation satisfying all of the group axioms except the axiom specifying the existence of inverses. Free objects in the category of monoids play a fundamental role in theoretical computer science where they model the behavior of machines (Turing machines, etc.). We shall encounter free algebras (i.e., polynomial algebras) and free modules in later chapters.

EXERCISES

1. Let F_1 and F_2 be free groups of finite rank. Prove that $F_1 \cong F_2$ if and only if they have the same rank. What facts do you need in order to extend your proof to infinite ranks (where the result is also true)?
2. Prove that if $|S| > 1$ then $F(S)$ is non-abelian.
3. Prove that the commutator subgroup of the free group on 2 generators is not finitely generated (in particular, subgroups of finitely generated groups need not be finitely generated).
4. Prove that every nonidentity element of a free group is of infinite order.
5. Establish a finite presentation for A_4 using 2 generators.
6. Establish a finite presentation for S_4 using 2 generators.
7. Prove that the following is a presentation for the quaternion group of order 8:

$$Q_8 = \langle a, b \mid a^2 = b^2, a^{-1}ba = b^{-1} \rangle.$$

8. Use presentations to find the orders of the automorphism groups of the groups $Z_2 \times Z_4$ and $Z_4 \times Z_4$.

9. Prove that $\text{Aut}(Q_8) \cong S_4$.

10. This exercise exhibits an automorphism of S_6 that is not inner (hence, together with Exercise 19 in Section 4.4 it shows that $|\text{Aut}(S_6) : \text{Inn}(S_6)| = 2$). Let $t'_1 = (1\ 2)(3\ 4)(5\ 6)$, $t'_2 = (1\ 4)(2\ 5)(3\ 6)$, $t'_3 = (1\ 3)(2\ 4)(5\ 6)$, $t'_4 = (1\ 2)(3\ 6)(4\ 5)$, and $t'_5 = (1\ 4)(2\ 3)(5\ 6)$. Show that $t'_1, \dots, t'_5$ satisfy the following relations:

$$(t'_i)^2 = 1 \text{ for all } i,$$

$$(t'_i t'_j)^2 = 1 \text{ for all } i \text{ and } j \text{ with } |i - j| \geq 2, \text{ and}$$

$$(t'_i t'_{i+1})^3 = 1 \text{ for all } i \in \{1, 2, 3, 4\}.$$

Deduce that $S_6 = \langle t'_1, \dots, t'_5 \rangle$ and that the map

$$(1\ 2) \mapsto t'_1, \quad (2\ 3) \mapsto t'_2, \quad (3\ 4) \mapsto t'_3, \quad (4\ 5) \mapsto t'_4, \quad (5\ 6) \mapsto t'_5$$

extends to an automorphism of S_6 (which is clearly not inner since it does not send transpositions to transpositions). [Use the presentation for S_6 described in Example 5.]

11. Let S be a set. The group with presentation (S, R) , where $R = \{[s, t] \mid s, t \in S\}$ is called the *free abelian group* on S — denote it by $A(S)$. Prove that $A(S)$ has the following universal property: if G is any abelian group and $\varphi : S \rightarrow G$ is any set map, then there is a unique group homomorphism $\Phi : A(S) \rightarrow G$ such that $\Phi|_S = \varphi$. Deduce that if A is a free abelian group on a set of cardinality n then

$$A \cong \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z} \quad (n \text{ factors}).$$

12. Let S be a set and let c be a positive integer. Formulate the notion of a *free nilpotent group* on S of nilpotence class c and prove it has the appropriate universal property with respect to nilpotent groups of class $\leq c$.
13. Prove that there cannot be a nilpotent group N generated by two elements with the property that *every* nilpotent group which is generated by two elements is a homomorphic image of N (i.e., the specification of the class c in the preceding problem was necessary).

Part II

RING THEORY

The theory of groups is concerned with general properties of certain objects having an algebraic structure defined by a single binary operation. The study of rings is concerned with objects possessing two binary operations (called addition and multiplication) related by the distributive laws. We first study analogues for the basic points of development in the structure theory of groups. In particular, we introduce subrings, quotient rings, ideals (which are the analogues of normal subgroups) and ring homomorphisms. We then focus on questions about general rings which arise naturally from the presence of two binary operations. Questions concerning multiplicative inverses lead to the notion of fields and eventually to the construction of some specific fields such as finite fields. The study of the arithmetic (divisibility, greatest common divisors, etc.) of rings such as the familiar ring of integers, $\mathbb{Z}$, leads to the notion of primes and unique factorizations in Chapter 8. The results of Chapters 7 and 8 are then applied to rings of polynomials in Chapter 9.

The basic theory of rings developed in Part II is the cornerstone for the remaining four parts of the book. The theory of ring actions (modules) comprises Part III of the book. There we shall see how the structure of rings is reflected in the structure of the objects on which they act and this will enable us to prove some powerful classification theorems. The structure theory of rings, in particular of polynomial rings, forms the basis in Part IV for the theory of fields and polynomial equations over fields. There the rich interplay among ring theory, field theory and group theory leads to many beautiful results on the structure of fields and the theory of roots of polynomials. Part V continues the study of rings and applications of ring theory to such topics as geometry and the theory of extensions. In Part VI the study of certain specific kinds of rings (group rings) and the objects (modules) on which they act again gives deep classification theorems whose consequences are then exploited to provide new results and insights into finite groups.

Introduction to Rings

7.1 BASIC DEFINITIONS AND EXAMPLES

Definition.

(1) A *ring* R is a set together with two binary operations $+$ and $\times$ (called addition and multiplication) satisfying the following axioms:

- (i) $(R, +)$ is an *abelian* group,
- (ii) $\times$ is associative : $(a \times b) \times c = a \times (b \times c)$ for all $a, b, c \in R$,
- (iii) the *distributive laws* hold in R : for all $a, b, c \in R$

$$(a + b) \times c = (a \times c) + (b \times c) \quad \text{and} \quad a \times (b + c) = (a \times b) + (a \times c).$$

(2) The ring R is *commutative* if multiplication is commutative.

(3) The ring R is said to have an *identity* (or *contain a 1*) if there is an element $1 \in R$ with

$$1 \times a = a \times 1 = a \quad \text{for all } a \in R.$$

We shall usually write simply ab rather than $a \times b$ for $a, b \in R$. The additive identity of R will always be denoted by 0 and the additive inverse of the ring element a will be denoted by $-a$.

The condition that R be a group under addition is a fairly natural one, but it may seem artificial to require that this group be *abelian*. One motivation for this is that if the ring R has a 1 , the commutativity under addition is *forced* by the distributive laws. To see this, compute the product $(1 + 1)(a + b)$ in two different ways, using the distributive laws (but not assuming that addition is commutative). One obtains

$$(1 + 1)(a + b) = 1(a + b) + 1(a + b) = 1a + 1b + 1a + 1b = a + b + a + b$$

and

$$(1 + 1)(a + b) = (1 + 1)a + (1 + 1)b = 1a + 1a + 1b + 1b = a + a + b + b.$$

Since R is a group under addition, this implies $b + a = a + b$, i.e., that R under addition is necessarily commutative.

Fields are one of the most important examples of rings. Note that their definition below is just another formulation of the one given in Section 1.4.

Definition. A ring R with identity 1 , where $1 \neq 0$, is called a *division ring* (or *skew field*) if every nonzero element $a \in R$ has a multiplicative inverse, i.e., there exists $b \in R$ such that $ab = ba = 1$. A commutative division ring is called a *field*.

More examples of rings follow.

Examples

- (1) The simplest examples of rings are the *trivial rings* obtained by taking R to be any commutative group (denoting the group operation by $+$) and defining the multiplication $\times$ on R by $a \times b = 0$ for all $a, b \in R$. It is easy to see that this multiplication defines a commutative ring. In particular, if $R = \{0\}$ is the trivial group, the resulting ring R is called the *zero ring*, denoted $R = 0$. Except for the zero ring, a trivial ring does not contain an identity ($R = 0$ is the only ring where $1 = 0$; we shall often exclude this ring by imposing the condition $1 \neq 0$). Although trivial rings have two binary operations, multiplication adds no new structure to the additive group and the theory of rings gives no information which could not already be obtained from (abelian) group theory.
- (2) The ring of integers, $\mathbb{Z}$, under the usual operations of addition and multiplication is a commutative ring with identity (the integer 1). The ring axioms (as with the additive group axioms) follow from the basic axioms for the system of natural numbers. Note that under *multiplication* $\mathbb{Z} - \{0\}$ is *not* a group (in fact, there are very few multiplicative inverses to elements in this ring). We shall come back to the question of these inverses shortly.
- (3) Similarly, the rational numbers, $\mathbb{Q}$, the real numbers, $\mathbb{R}$, and the complex numbers, $\mathbb{C}$, are commutative rings with identity (in fact they are fields). The ring axioms for each of these follow ultimately from the ring axioms for $\mathbb{Z}$. We shall verify this when we construct $\mathbb{Q}$ from $\mathbb{Z}$ (Section 7.5) and $\mathbb{C}$ from $\mathbb{R}$ (Example 1, Section 13.1); both of these constructions will be special cases of more general processes. The construction of $\mathbb{R}$ from $\mathbb{Q}$ (and subsequent verification of the ring axioms) is carried out in basic analysis texts.
- (4) The quotient group $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with identity (the element $\bar{1}$) under the operations of addition and multiplication of residue classes (frequently referred to as “modular arithmetic”). We saw that the additive abelian group axioms followed from the general principles of the theory of quotient groups (indeed this was the prototypical quotient group). We shall shortly prove that the remaining ring axioms (in particular, the fact that multiplication of residue classes is well defined) follow analogously from the general theory of quotient rings.

In all of the examples so far the rings have been commutative. Historically, one of the first noncommutative rings was discovered in 1843 by Sir William Rowan Hamilton (1805–1865). This ring, which is a division ring, was extremely influential in the subsequent development of mathematics and it continues to play an important role in certain areas of mathematics and physics.

- (5) (The *real*) *Hamilton Quaternions*) Let $\mathbb{H}$ be the collection of elements of the form $a + bi + cj + dk$ where $a, b, c, d \in \mathbb{R}$ are real numbers (loosely, “polynomials in i, j, k with real coefficients”) where addition is defined “componentwise” by

$$(a+bi+cj+dk) + (a'+b'i+c'j+d'k) = (a+a') + (b+b')i + (c+c')j + (d+d')k$$

and multiplication is defined by expanding $(a + bi + cj + dk)(a' + b'i + c'j + d'k)$ using the distributive law (being careful about the order of terms) and simplifying

using the relations

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j$$

(where the real number coefficients commute with i , j and k). For example,

$$\begin{aligned}(1+i+2j)(j+k) &= 1(j+k) + i(j+k) + 2j(j+k) = j + k + ij + ik + 2j^2 + 2jk \\ &= j + k + k + (-j) + 2(-1) + 2(i) = -2 + 2i + 2k.\end{aligned}$$

The fact that $\mathbb{H}$ is a ring may be proved by a straightforward, albeit lengthy, check of the axioms (associativity of multiplication is particularly tedious). The Hamilton Quaternions are a noncommutative ring with identity ($1 = 1+0i+0j+0k$). Similarly, one can define the ring of *rational* Hamilton Quaternions by taking a, b, c, d to be rational numbers above. Both the real and rational Hamilton Quaternions are *division rings*, where inverses of nonzero elements are given by

$$(a + bi + cj + dk)^{-1} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}.$$

- (6) One important class of rings is obtained by considering rings of functions. Let X be any nonempty set and let A be any ring. The collection, R , of all (set) functions $f : X \rightarrow A$ is a ring under the usual definition of pointwise addition and multiplication of functions: $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$. Each ring axiom for R follows directly from the corresponding axiom for A . The ring R is commutative if and only if A is commutative and R has a 1 if and only if A has a 1 (in which case the 1 of R is necessarily the constant function 1 on X).

If X and A have more structure, we may form other rings of functions which respect those structures. For instance, if A is the ring of real numbers $\mathbb{R}$ and X is the closed interval $[0, 1]$ in $\mathbb{R}$ we may form the ring of all *continuous* functions from $[0, 1]$ to $\mathbb{R}$ (here we need basic limit theorems to guarantee that sums and products of continuous functions are continuous) — this is a commutative ring with 1.

- (7) An example of a ring which does not have an identity is the ring $2\mathbb{Z}$ of even integers under usual addition and multiplication of integers (the sum and product of even integers is an even integer).

Another example which arises naturally in analysis is constructed as follows. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to have *compact support* if there are real numbers a, b (depending on f) such that $f(x) = 0$ for all $x \notin [a, b]$ (i.e., f is zero outside some bounded interval). The set of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with compact support is a commutative ring without identity (since an identity could not have compact support). Similarly, the set of all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with compact support is a commutative ring without identity.

In the next section we give three important ways of constructing “larger” rings from a given ring (analogous to Example 6 above) and thus greatly expand our list of examples. Before doing so we mention some basic properties of arbitrary rings. The ring $\mathbb{Z}$ is a good example to keep in mind, although this ring has a good deal more algebraic structure than a general ring (for example, it is commutative and has an identity). Nonetheless, its basic arithmetic holds for general rings as the following result shows.

Proposition 1. Let R be a ring. Then

- (1) $0a = a0 = 0$ for all $a \in R$.
- (2) $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$ (recall $-a$ is the additive inverse of a).
- (3) $(-a)(-b) = ab$ for all $a, b \in R$.
- (4) if R has an identity 1 , then the identity is unique and $-a = (-1)a$.

Proof: These all follow from the distributive laws and cancellation in the additive group R . For example, (1) follows from $0a = (0 + 0)a = 0a + 0a$. The equality $(-a)b = -(ab)$ in (2) follows from $ab + (-a)b = (a + (-a))b = 0b = 0$. The rest follow similarly and are left to the reader.

This proposition shows that because of the distributive laws the additive and multiplicative structures of a ring behave well with respect to one another, just as in the familiar example of the integers.

Unlike the integers, however, general rings may possess many elements that have multiplicative inverses or may have nonzero elements a and b whose product is zero. These two properties of elements, which relate to the multiplicative structure of a ring, are given special names.

Definition. Let R be a ring.

- (1) A nonzero element a of R is called a *zero divisor* if there is a nonzero element b in R such that either $ab = 0$ or $ba = 0$.
- (2) Assume R has an identity $1 \neq 0$. An element u of R is called a *unit* in R if there is some v in R such that $uv = vu = 1$. The set of units in R is denoted $R^\times$.

It is easy to see that the units in a ring R form a group under multiplication so $R^\times$ will be referred to as the *group of units* of R . In this terminology a *field* is a commutative ring F with identity $1 \neq 0$ in which every nonzero element is a unit, i.e., $F^\times = F - \{0\}$.

Observe that a zero divisor can never be a unit. Suppose for example that a is a unit in R and that $ab = 0$ for some nonzero b in R . Then $va = 1$ for some $v \in R$, so $b = 1b = (va)b = v(ab) = v0 = 0$, a contradiction. Similarly, if $ba = 0$ for some nonzero b then a cannot be a unit.

This shows in particular that fields contain no zero divisors.

Examples

- (1) The ring $\mathbb{Z}$ of integers has no zero divisors and its only units are ± 1 , i.e., $\mathbb{Z}^\times = \{\pm 1\}$. Note that every nonzero integer has an inverse in the larger ring $\mathbb{Q}$, so the property of being a unit depends on the ring in which an element is viewed.
- (2) Let n be an integer ≥ 2 . In the ring $\mathbb{Z}/n\mathbb{Z}$ the elements $\bar{u}$ for which u and n are relatively prime are units (we shall prove this in the next chapter). Thus our use of the notation $(\mathbb{Z}/n\mathbb{Z})^\times$ is consistent with the definition of the group of units in an arbitrary ring.

If, on the other hand, a is a nonzero integer and a is not relatively prime to n then we show that $\bar{a}$ is a zero divisor in $\mathbb{Z}/n\mathbb{Z}$. To see this let d be the g.c.d. of a and n and let $b = \frac{n}{d}$. By assumption $d > 1$ so $0 < b < n$, i.e., $\bar{b} \neq \bar{0}$. But by construction n